Chapter 4
Physics of Bipolar Transistors

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Chapter 4 Physics of Bipolar Transistors

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- 4.2 Structure of Bipolar Transistor
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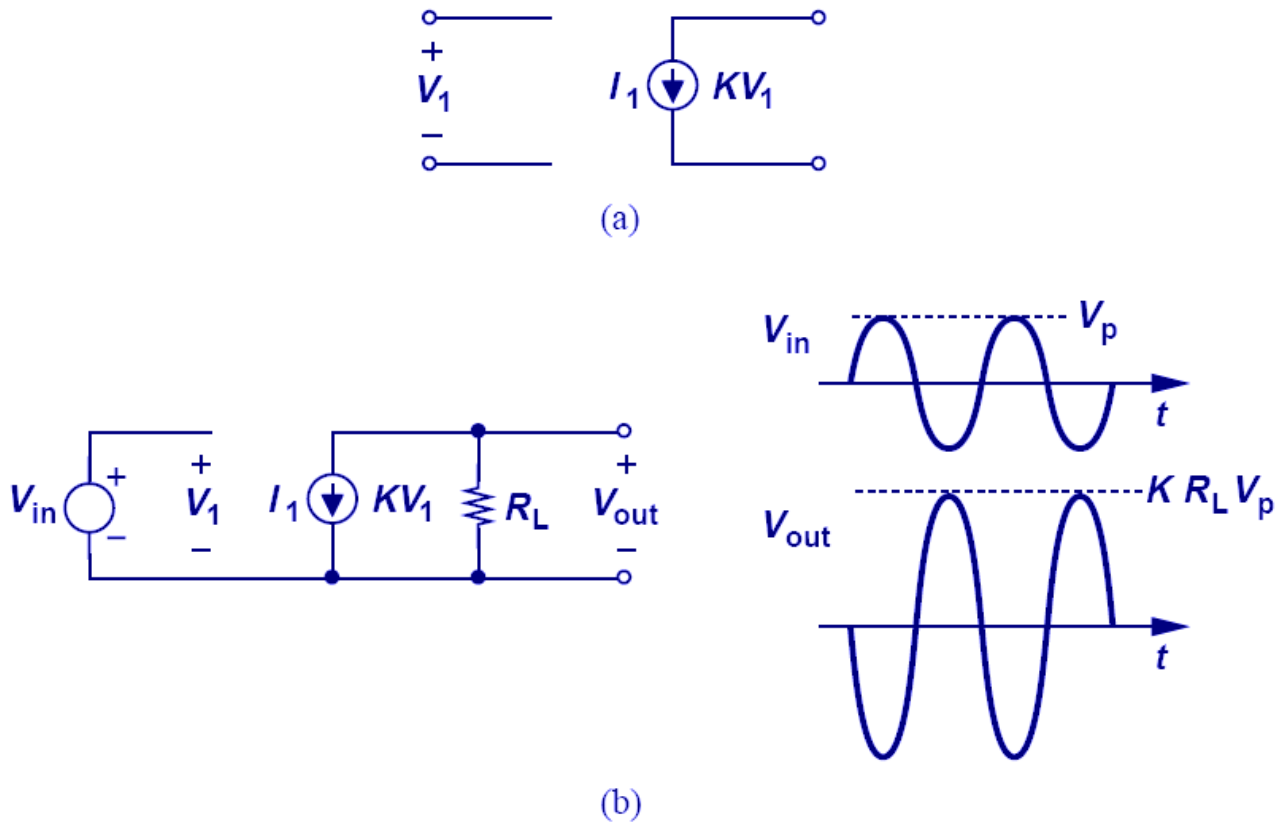
Bipolar Transistor

- In the chapter, we will study the physics of bipolar transistor and derive large and small signal models.



Voltage-Dependent Current Source

- A voltage-dependent current source can act as an amplifier.
- If KR_L is greater than 1, then the signal is amplified.



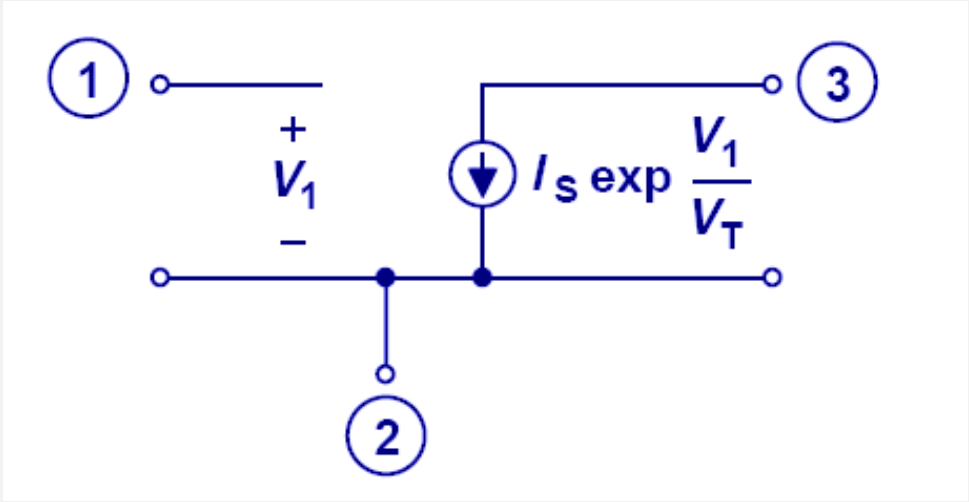
(a) Voltage-dependent current source

(b) simple amplifier

$$A_V = \frac{V_{out}}{V_{in}} = -KR_L$$

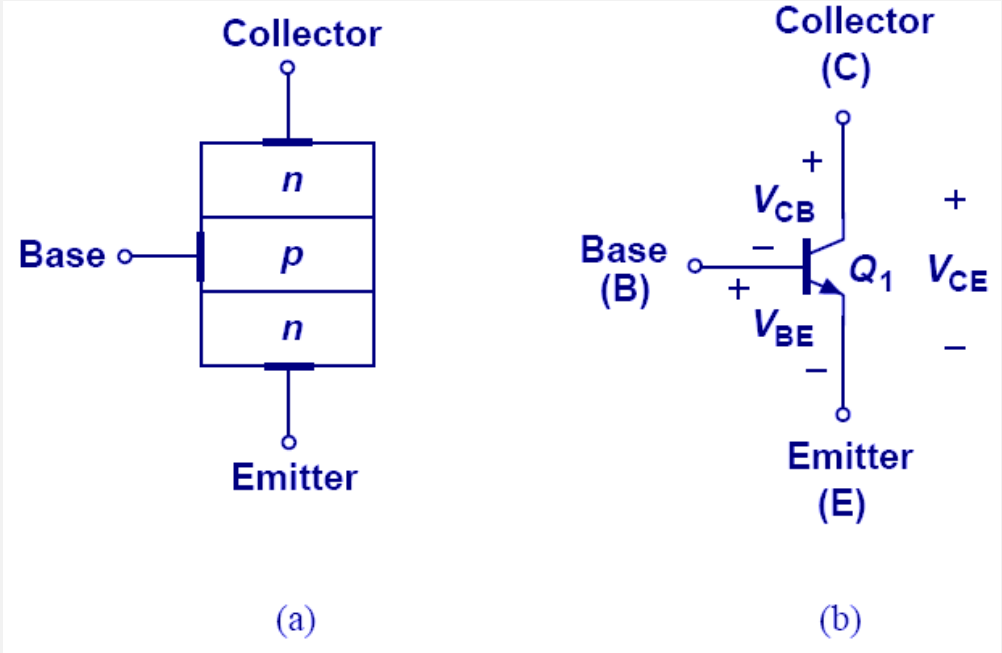
Exponential Voltage-Dependent Current Source

- A voltage-controlled current source can indeed provide signal amplification.
- Bipolar transistors are an example of such current sources and can ideally be modeled as below.



Structure and Symbol of Bipolar Transistor

- Bipolar transistor can be thought of as a sandwich of three doped Si regions. The outer two regions are doped with the same polarity, while the middle region is doped with opposite polarity.
- The emitter “emits” charge carriers and the collector “collects” them while the base controls the number of carriers.



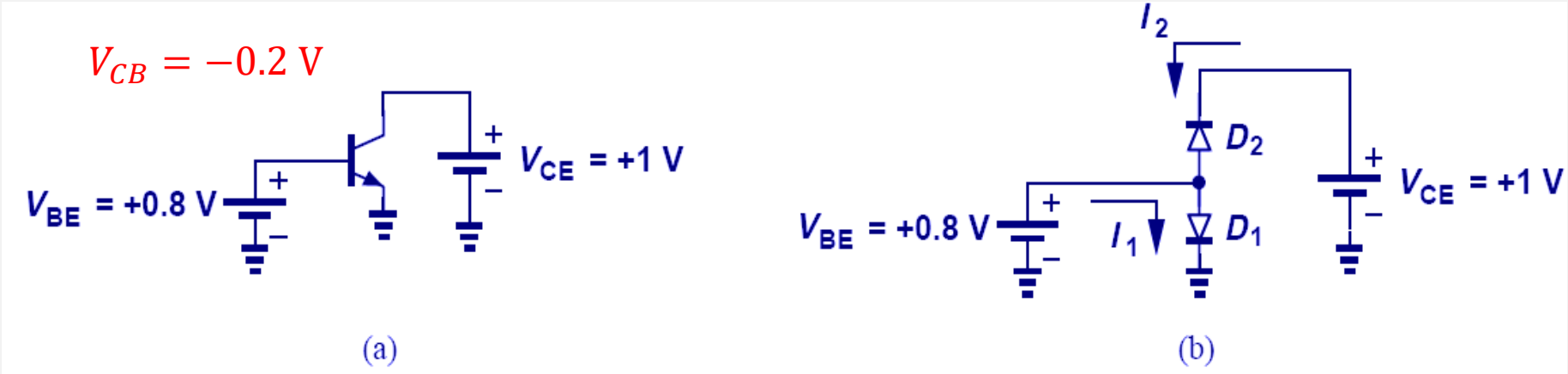
(a) Structure and (b) circuit symbol of an “*npn*” bipolar transistor.

Operation Modes of Bipolar Transistor

- There are 2^2 possibilities for the terminal voltages of a bipolar transistor, depending on the bias conditions of the junctions: V_{BE}, V_{CB}, and V_{CE}.
- Fortunately, three combinations (modes) are practical.
  (1) Active: $V_{BE} > 0$, $V_{CB} < 0$
  (2) Cutoff: $V_{BE} < 0$, $V_{CB} < 0$
  (3) Saturation: $V_{BE} > 0$, $V_{CB} > 0$

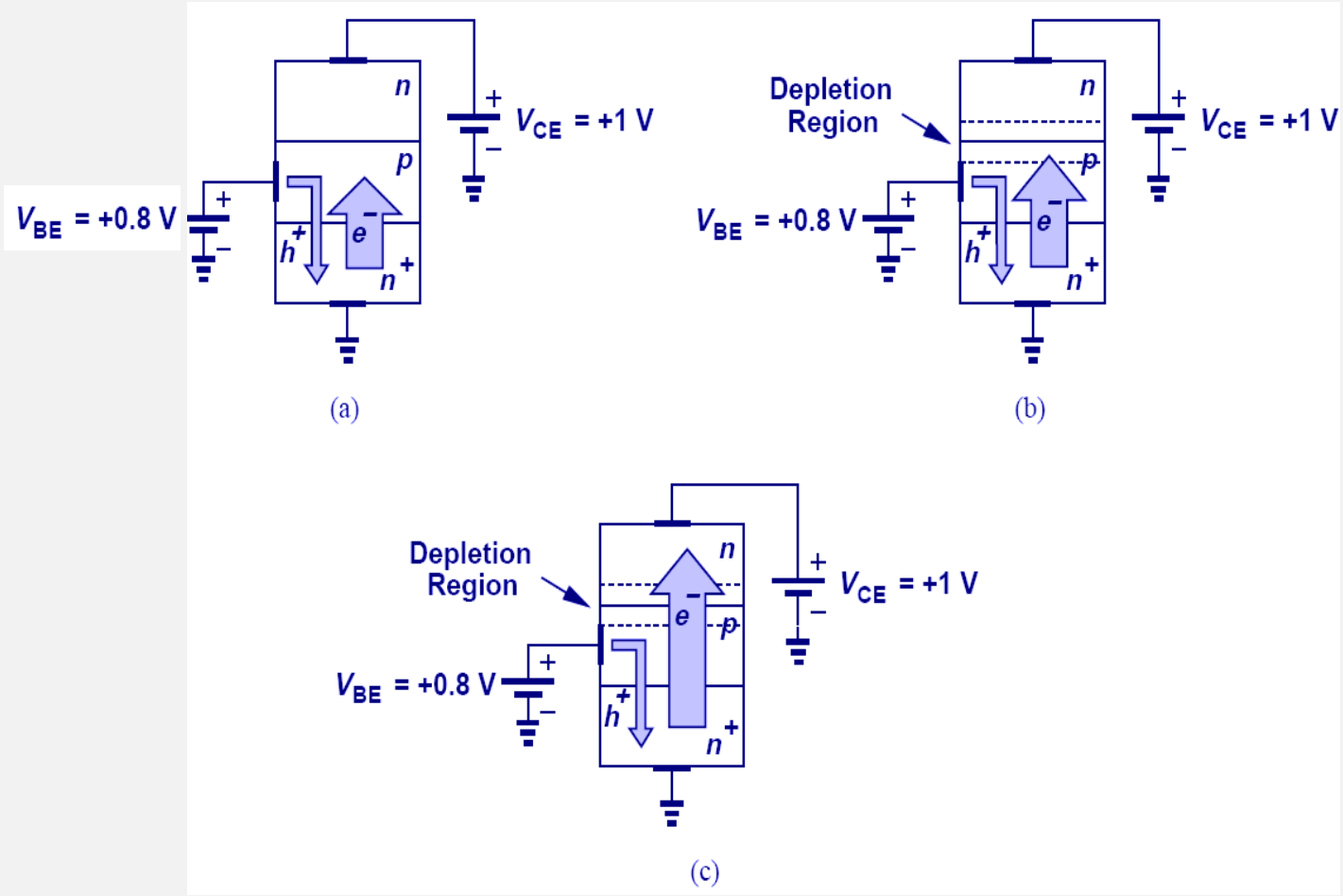
Operation of Bipolar Transistor in Active Mode

- Forward active region: $V_{BE} > 0$, $V_{CB} < 0$
- Wrong model: the transistor cannot be modeled as merely two back-to back diodes
  - Base region is very thin (~100 Å)
  - High doping level in emitter (n^+): the emitter doping level is much greater than the base



(b) presents a wrong way of modeling (a)

Carrier Flow of Bipolar Transistor in Active Mode



In the active mode,

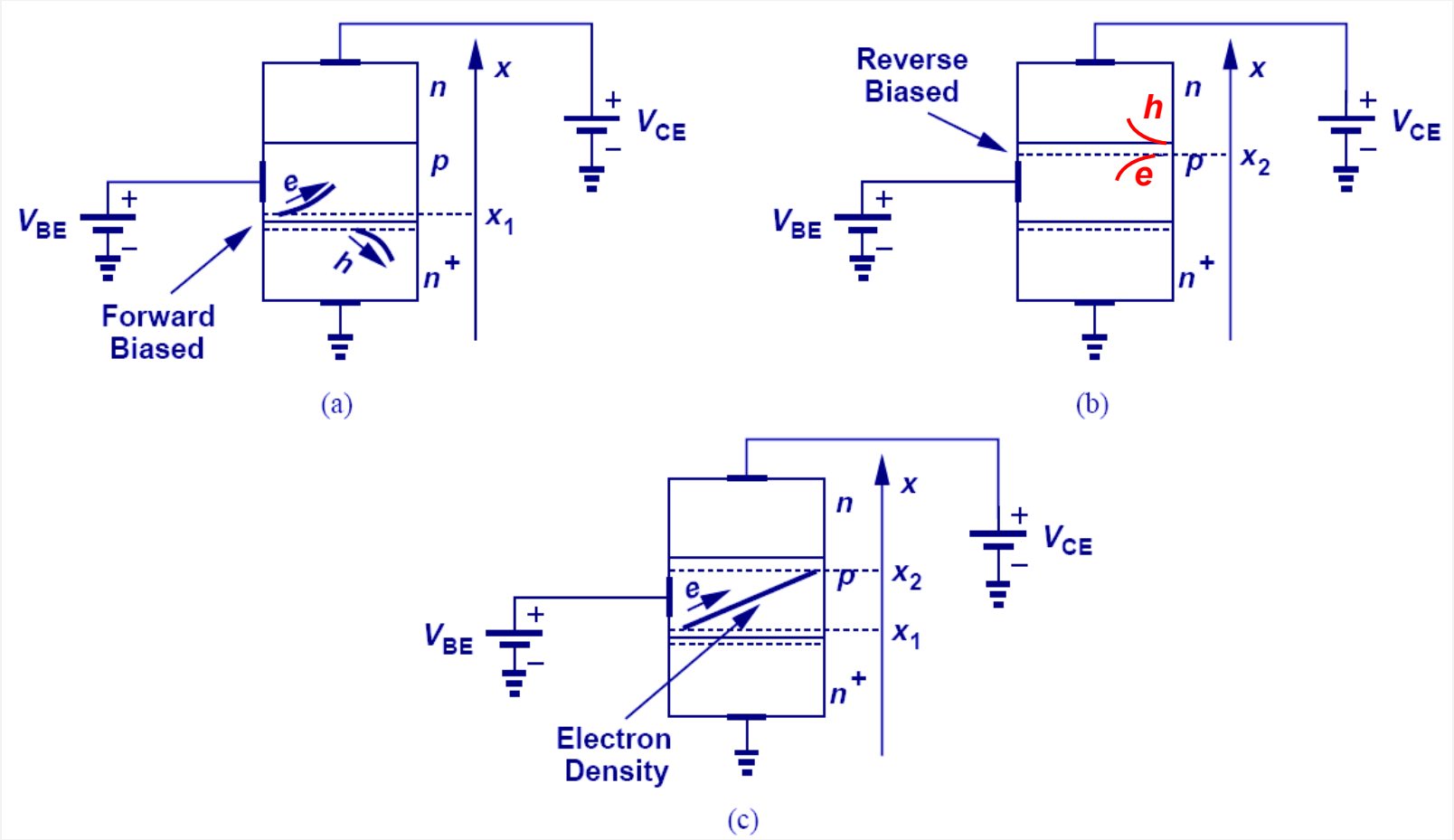
(a) an npn bipolar transistor carries a large number of electrons (e^-) from the emitter, while drawing a small current of holes (h^+) through the base terminal. ($e^- \gg h^+$)

(b) the base region is thin, most of the electrons reach the edge of the collector-base depletion region.

(c) nearly all electrons pass through collector junction. ($I_c \approx I_E$)

Carriers Profile in Base

- Forward between BE ($V_{BE} > 0$) and reverse bias between CB ($V_{CB} < 0$)



Collector Current

As the base-emitter junction is in forward bias, the electron concentration at boundary x_1 is

$$n(x_1) = N_E e^{-\left(\frac{V_0 - V_{BE}}{V_T}\right)} = N_E e^{-\frac{V_0}{V_T}} \times e^{\frac{V_{BE}}{V_T}}$$

where N_E is doping level in the emitter. The excess concentration of electron is defined as

$$\Delta n(x_1) \triangleq n(x_1) - n_{p,e} = N_E e^{-\frac{V_0}{V_T}} \times e^{\frac{V_{BE}}{V_T}} - N_E e^{-\frac{V_0}{V_T}}$$

$$= N_E e^{-V_0/V_T}\left(e^{V_{BE}/V_T} - 1\right) \qquad (1)$$

The flow of electrons into the collector can be approximated as

$$J_n = qD_n \frac{dn}{dx} = qD_n \frac{n(x_1) - n(x_2)}{W_B} \approx qD_n \frac{n(x_1) - n_{p,e}}{W_B} = qD_n \frac{\Delta n(x_1)}{W_B} \qquad (2)$$

where W_B is the width of the base region.

Substitute (1) into (2),

$$J_n = \frac{qD_n}{W_B} N_E e^{-V_0/V_T} \left(e^{V_{BE}/V_T} - 1\right)$$

$$= \frac{qD_n}{W_B} \left(\frac{{n_i}^2}{N_B}\right) \left(e^{V_{BE}/V_T} - 1\right) \quad \left(\because e^{\frac{V_0}{V_T}} = \frac{N_E N_B}{{n_i}^2} \Rightarrow N_E e^{-V_0/V_T} = \frac{{n_i}^2}{N_B}\right)$$

where N_B is doping level in the base. The collector current can be approximated as

$$I_C = A_E\left(J_n - J_p\right) \approx A_E J_n \qquad \left(J_n \gg J_p\right)$$

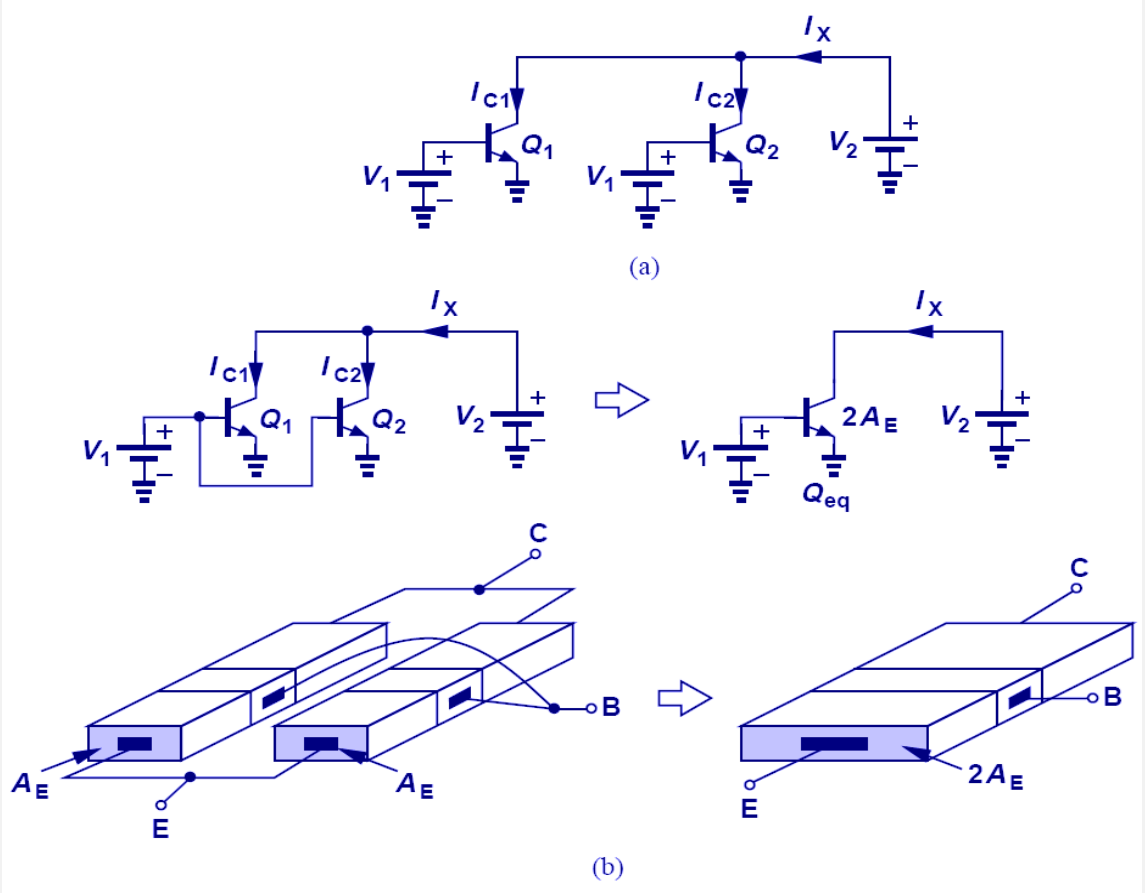
$$= \frac{A_E q D_n n_i^2}{N_B W_B}\left(e^{V_{BE}/V_T} - 1\right)$$

where A_E is cross section area of the emitter. Assume $e^{V_{BE}/V_T} \gg 1$,

$$I_C = I_S e^{V_{BE}/V_T}, \text{ where } I_S = \frac{A_E q D_n n_i^2}{N_B W_B}.$$

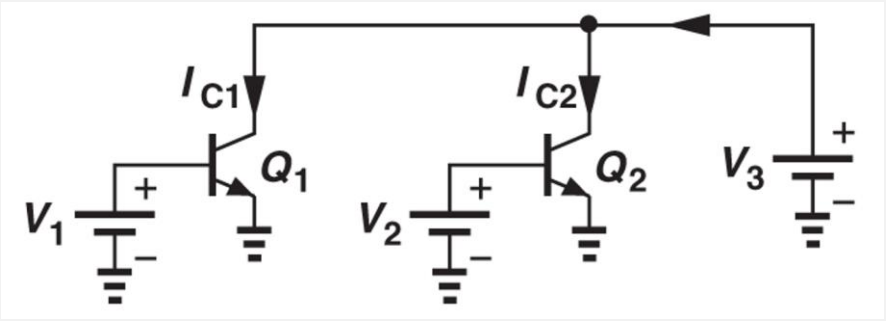
Parallel Combination of Transistors

- When two transistors are put in parallel and experience the same potential across all three terminals, they can be thought of as a single transistor with twice the emitter area.



Example 4-3

- In the following circuit, Q_1 and Q_2 are identical and operate in the active mode. Determine $V_1 - V_2$ such that $I_{C1} = 10I_{C2}$ at room temperature.



Solution:

$$I_{C1} = 10I_{C2} \Rightarrow I_{C1}/I_{C2} = 10$$

$$\frac{I_S e^{V_1/V_T}}{I_S e^{V_2/V_T}} = 10$$

$$e^{(V_1 - V_2)/V_T} = 10$$

$$V_1 - V_2 = V_T \ln 10$$

$$= 26\ \text{mV} \times \ln 10 = 59.9\ \text{mV}$$

Example 4-4

- Typical discrete bipolar transistors have a large area, e.g., $500\,\mu\text{m} \times 500\,\mu\text{m}$, whereas modern integrated devices may have an area as small as $0.5\,\mu\text{m} \times 0.2\,\mu\text{m}$. Assuming other device parameters are identical, determine the difference between the base-emitter voltage of two such transistors for equal collector currents.

Solution:

$$I_C = I_S e^{V_{BE}/V_T} \implies V_{BE} = V_T \ln \frac{I_C}{I_S}$$

$$V_{BE,dis} \triangleq V_T \ln \frac{I_C}{I_{S1}}, \qquad V_{BE,IC} \triangleq V_T \ln \frac{I_C}{I_{S2}}$$

$$I_S = \frac{A_E q D_n n_i^2}{N_E W_B} \implies I_S \propto A_E \text{ for identical other device parameters}$$

$$\begin{aligned} V_{BE,IC} - V_{BE,dis} &= V_T \ln \frac{I_C}{I_{S1}} - V_T \ln \frac{I_C}{I_{S2}} \\ &= V_T \ln \frac{I_{S2}}{I_{S1}} = V_T \ln \frac{A_{E2}}{A_{E1}} \\ &= 26\,\text{mV} \times \ln \frac{500\,\mu\text{m} \times 500\,\mu\text{m}}{0.5\,\mu\text{m} \times 0.2\,\mu\text{m}} = 383\,\text{mV} \end{aligned}$$

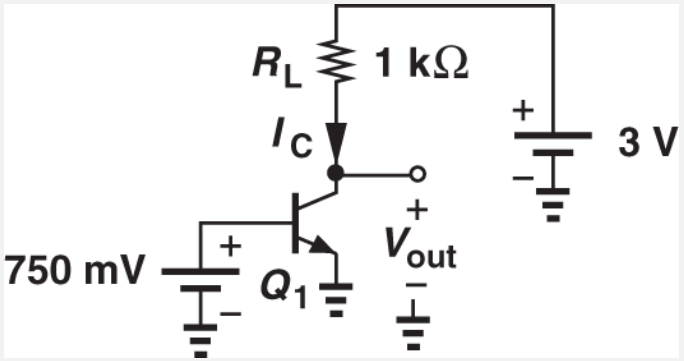
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Note:

- ✓ In practice, however, $V_{BE,IC} - V_{BE,dis}$ falls in the range of 100 to 150 mV because of differences in the base width and other parameters.
- ✓ The key point here is that $V_{BE,IC}$ = 800 mV is a reasonable approximation for integrated transistors and should be lowered to about $V_{BE,dis}$ = 700 mV for discrete devices.

Example 4-5 Simple stage with biasing

- Determine the output voltage if $I_S = 5 \times 10^{-16}$ A.



Solution:

$$I_C = I_S e^{V_{BE}/V_T}$$

$$I_C = 5 \times 10^{-16} e^{750/26} = 1.69 \text{ mA}$$

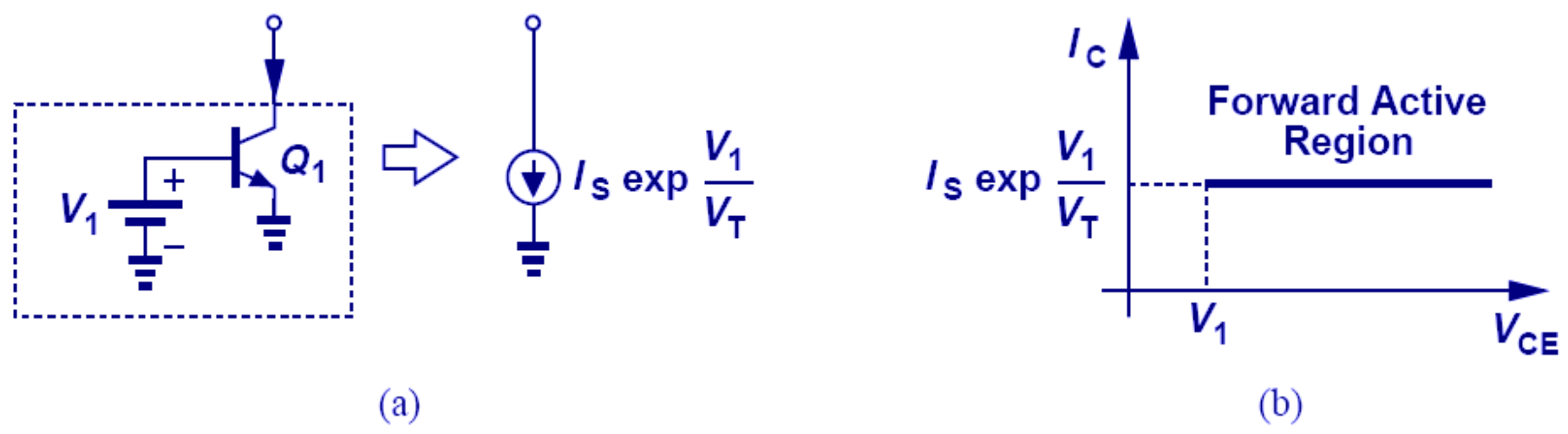
$$V_{CC} = I_C R_L + V_{out}$$

$$3 = 1.69 \text{ mA} \times 1 \text{ k}\Omega + V_{out}$$

$$V_{out} = 1.31 \text{ V}$$

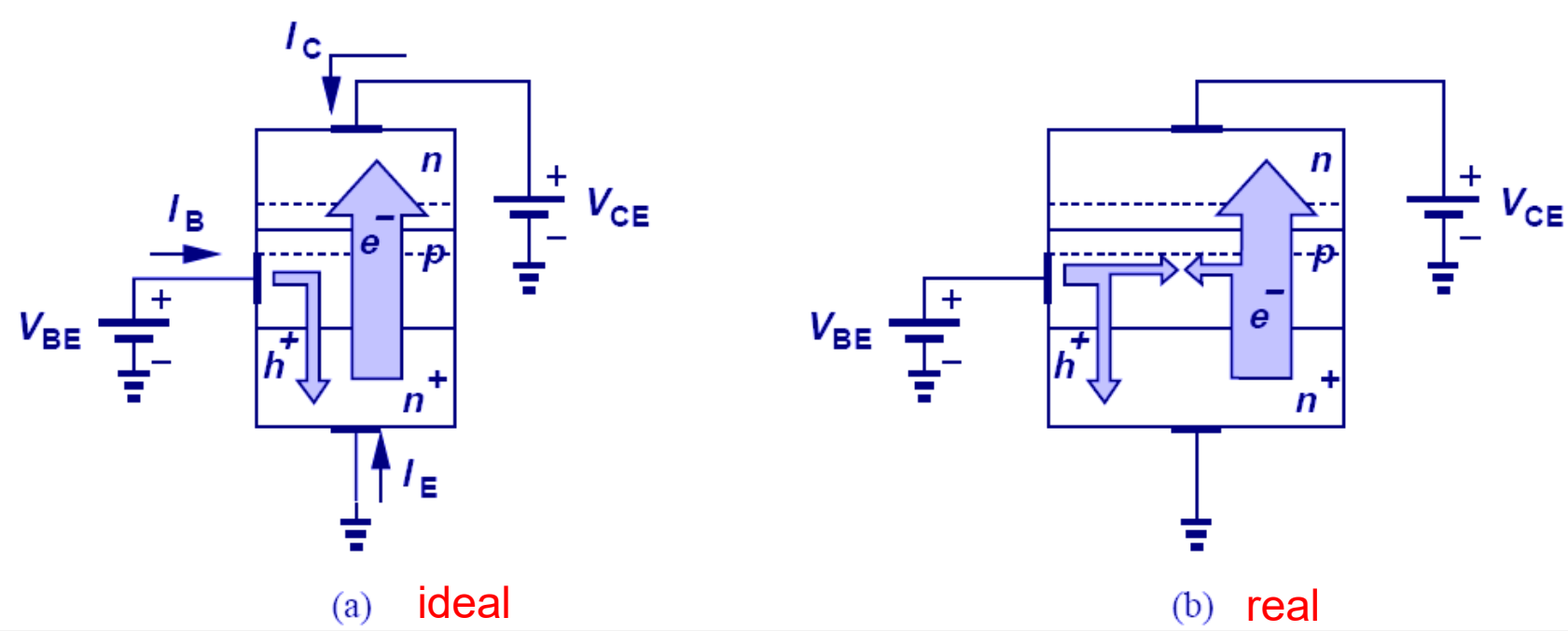
Constant Current Source

- Previous example shows that the collector current does not depend on the collector to emitter voltage (so long as the device remains in the active mode).
- This property allows the transistor to behave as a constant current source when its base-emitter voltage is fixed.



Base Current

- Base current consists of two components: 1) Reverse injection of holes into the emitter and 2) recombination of holes with electrons coming from the emitter.



$$I_C = \beta I_B$$

- $\beta = I_C / I_B$: Base current I_B bears a *constant* ratio given by the doping levels and other parameters.
  - ✓ As an example, for every 200 electrons injected by the emitter, one hole must be supplied by the base.
  - ✓ Electrons injected by the emitter travel through the base, some may "recombine" with the holes, in essence, some electrons and holes are "wasted" as a result of recombination.
- β is called the "current gain" of the transistor because it shows how much the base current is "amplified."
  - ✓ Depending on the device structure, the of npn transistors typically ranges from 50 to 200.

Emitter Current

- Applying KCL to the transistor, we can easily find the emitter current.

$$I_E = I_C + I_B$$
$$= I_C\left(1 + \frac{1}{\beta}\right)$$

where $\beta = \frac{I_C}{I_B}$.

Summary of Currents

- Given device parameters I_S and $\beta = I_C/I_B$,

$$I_C = I_S \exp \frac{V_{BE}}{V_T}$$
$$I_B = \frac{1}{\beta} I_S \exp \frac{V_{BE}}{V_T}$$
$$I_E = \frac{\beta + 1}{\beta} I_S \exp \frac{V_{BE}}{V_T}$$

- It is sometimes useful to write

$$I_C = \alpha I_E, \text{ where } \alpha = \frac{\beta}{\beta+1}.$$

- For $\beta = 100$ 100, $\alpha = 0.99$, suggesting that $\alpha \approx 1$ and $I_C \approx I_E$ are reasonable approximations. That means the collector and emitter currents are approximately equal.

Example 4-6

- A bipolar transistor having $I_S = 5 \times 10^{-16}$ A is biased in the forward active region with V_{BE} = 750 mV. If the current gain varies from 50 to 200 due to manufacturing variations, calculate the minimum and maximum terminal currents of the device.
- **Solution**:

$$I_C = I_S e^{V_{BE}/V_T} = 5 \times 10^{-16} \cdot e^{750/26} = 1.685 \text{ mA}$$

$$I_B = \frac{I_C}{\beta},\ 50 < \beta < 200$$

$$\frac{I_C}{200} < I_B < \frac{I_C}{50} \implies 8.43\ \mu\text{A} < I_B < 33.7\ \mu\text{A}$$

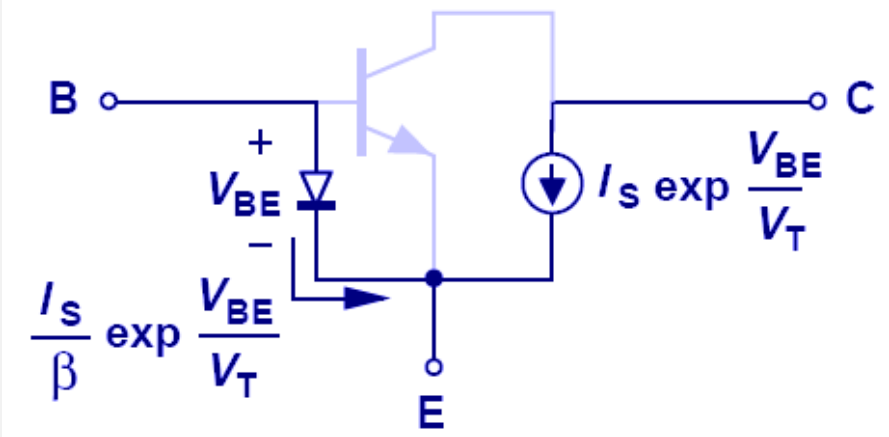
$$I_E = \left(1 + \frac{1}{\beta}\right) I_C$$

$$I_C\left(1 + \frac{1}{200}\right) < I_E < I_C\left(1 + \frac{1}{50}\right) \implies 1.005 I_C < I_E < 1.02 I_C$$

$$1.69 \text{ mA} < I_E < 1.719 \text{ mA}$$

Bipolar Transistor Large-Signal Model

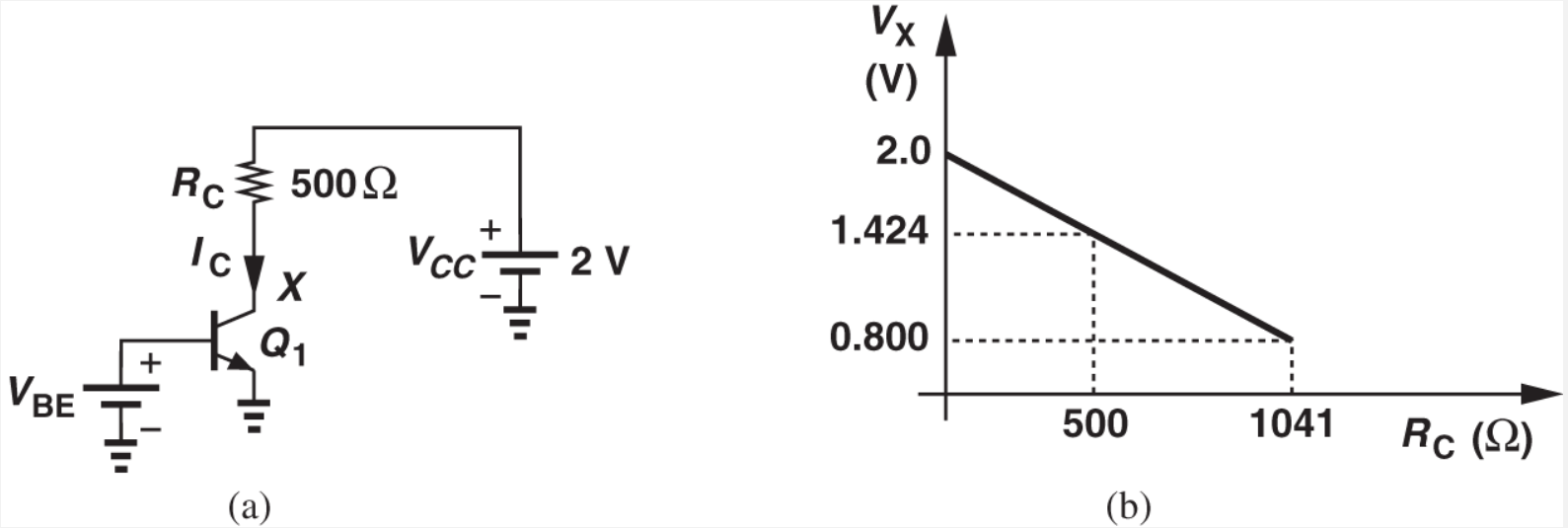
- A diode is placed between base and emitter and a voltage controlled current source is placed between the collector and emitter.
- $V_{BE} \longrightarrow I_C \longrightarrow I_B \longrightarrow I_E$: The sequence of current and voltage dependencies begins with the base-emitter voltage controlling the collector current, which is proportional to the base current, with their sum flowing through the emitter.



Large-signal model of bipolar transistor in active region.

Example 4-7

- Consider the circuit shown in below, where $I_{S,Q1} = 5 \times 10^{-17}$ A and V_{BE} = 800 mV. Assume β = 100. (a) Determine the transistor terminal currents and voltages and verify that the device indeed operates in the active mode, (b) Determine the maximum value of R_C that permits operation in the active mode.



(a) Simple stage with biasing, (b) variation of collector voltage as a function of collector resistance.

- **Solution**:

(a)

$$I_C = I_S \exp\frac{V_{BE}}{V_T} = 5 \times 10^{-17} \exp\frac{800\ \text{mV}}{26\ \text{mV}} = 1.153\ \text{mA}$$

$$I_B = \frac{I_C}{\beta} = \frac{1.153\ \text{mA}}{100} = 11.53\ \mu\text{A}$$

$$I_E = \left(1 + \frac{1}{\beta}\right) I_C = \left(1 + \frac{1}{100}\right) \times 1.153\ \text{mA} = 1.165\ \text{mA}$$

$$V_{CC} = I_C R_C + V_X$$
$$2\ \text{V} = 1.153\ \text{mA} \times 500\ \Omega + V_X$$
$$V_X = 1.424\ \text{V}$$
$$V_{CB} = V_X - V_B = 1.424 - 0.8 = 0.624 > 0$$

The junction is reverse-biased and the transistor operates in the active mode.

(b)

As $V_X \rightarrow V_{BE} = 800\ \text{mV}$, the device approaches the “edge” of the forward active region.

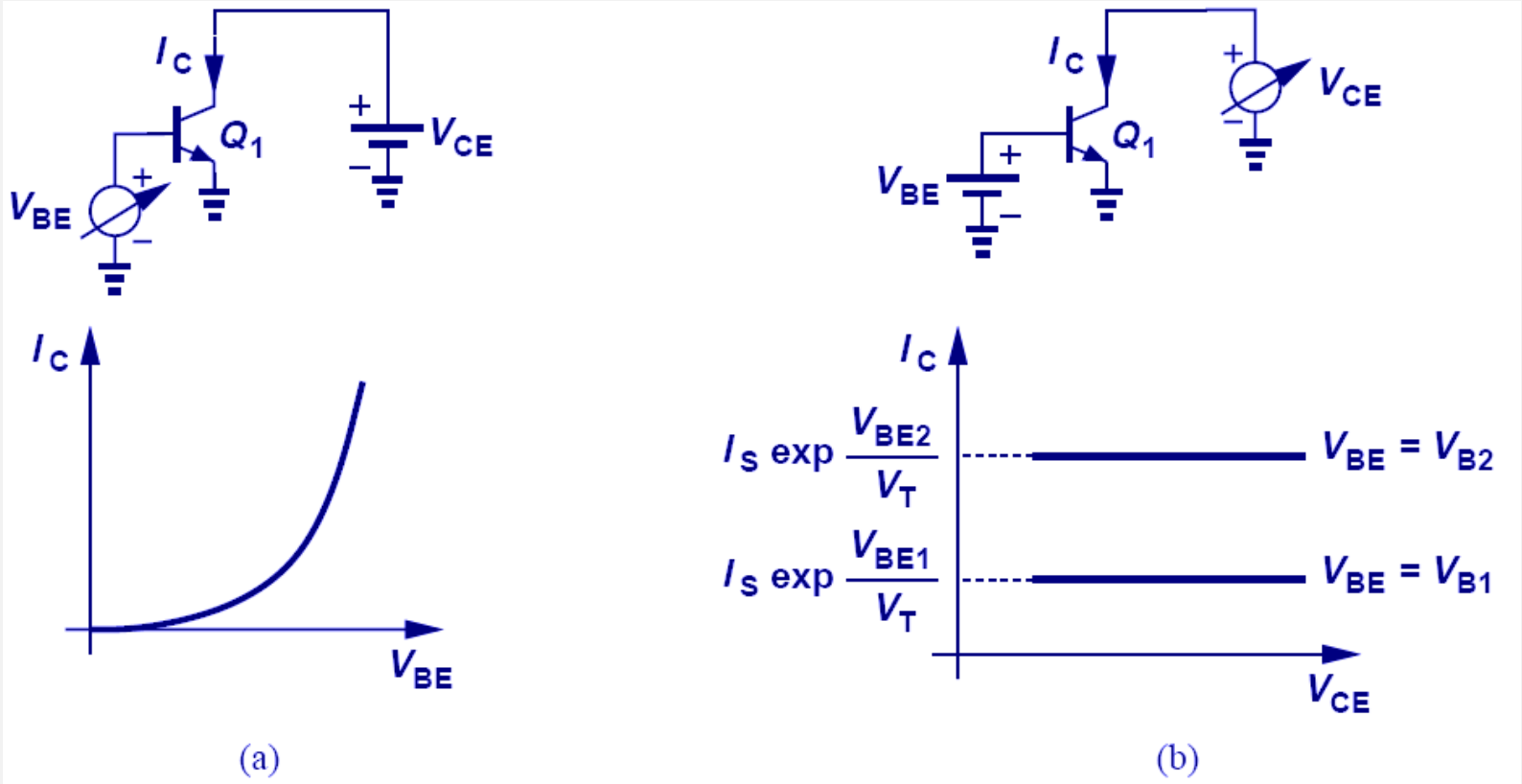
$$R_C = \frac{V_{CC} - V_X}{I_C} = \frac{2 - 0.8}{1.153\ \text{mA}} = 1041\ \Omega$$

Thus, the maximum allowable value of the collector resistance is $1041\ \Omega$.

- ✓ As R_C increases, V_X drops and eventually forward biases the collector-base junction. This will force the transistor out of forward active region.
- ✓ Therefore, there exists a maximum tolerable collector resistance.

I-V Characteristics of Bipolar Transistor

- If the device remains in the active mode ($V_{CE} > V_{BE}$), I_C is independent of V_{CE}.



Collector current as a function of (a) base-emitter voltage and (b) collector-emitter voltage.

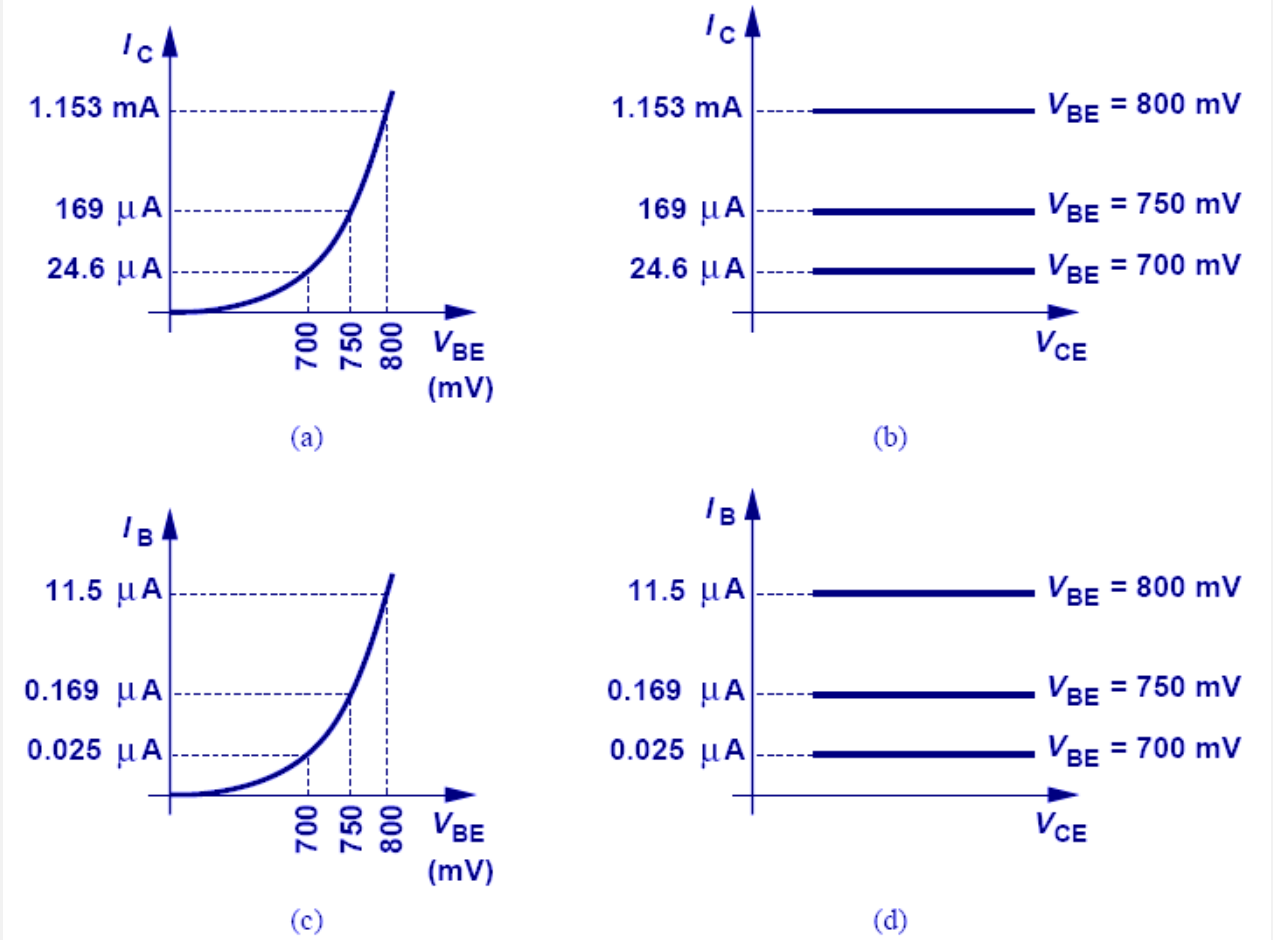
Example 4-8

- For a bipolar transistor, $I_S = 5 \times 10^{-17}$ A and β = 100. Construct the $I_C - V_{BE}$, $I_C - V_{CE}$, $I_B - V_{BE}$, and $I_B - V_{CE}$ characteristics.
- **Solution**:

$$I_C = I_S \exp \frac{V_{BE}}{V_T}$$

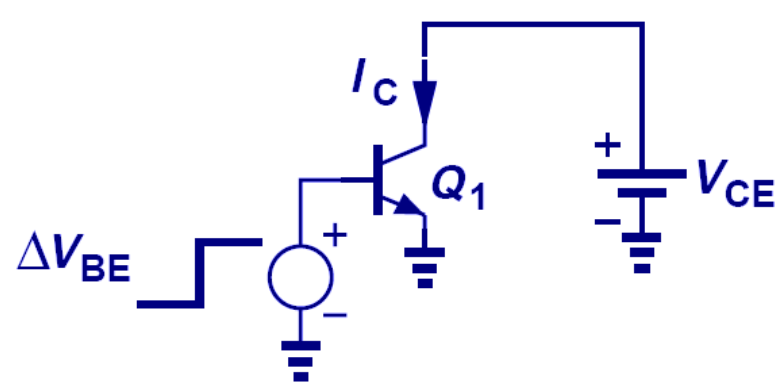
$$V_{BE1} = 700 \text{ mA} \Longrightarrow I_{C1} = 24.6\ \mu\text{A}$$
$$V_{BE2} = 750 \text{ mA} \Longrightarrow I_{C2} = 169\ \mu\text{A}$$
$$V_{BE3} = 800 \text{ mA} \Longrightarrow I_{C3} = 1.153 \text{ mA}$$

$$I_B = \frac{I_C}{\beta}$$



Concept of Transconductance

- How is the *performance* of such a device quantified? In other words, what is the measure of the "goodness" of a voltage-dependent current source?
- Transconductance (g_m) shows a measure of how well the transistor converts voltage to current.



Test circuit for measurement of g_m

$$g_m \triangleq \frac{dI_C}{dV_{BE}}$$

$$g_m = \frac{d}{dV_{BE}}\left(I_S \exp\frac{V_{BE}}{V_T}\right)$$

$$g_m = \frac{1}{V_T} I_S \exp\frac{V_{BE}}{V_T}$$

$$g_m = \frac{I_C}{V_T}$$

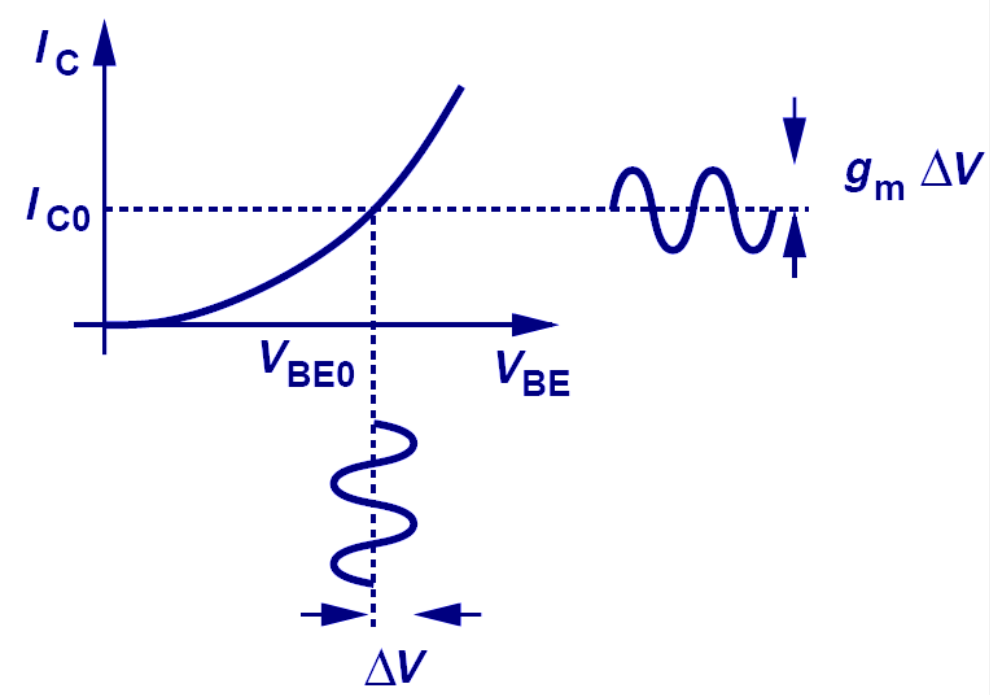
Transconductance

- The transconductance may be expressed in Ω^{-1} or “siemens,” (S).
- For example, if I_C = 1 mA, then with V_T = 26 mV, we have

$$\begin{aligned} g_m &= 0.0385\ \Omega^{-1} \\ &= 0.0385\ \mathrm{S} \\ &= 38.5\ \mathrm{mS} \end{aligned}$$

Visualization of Transconductance

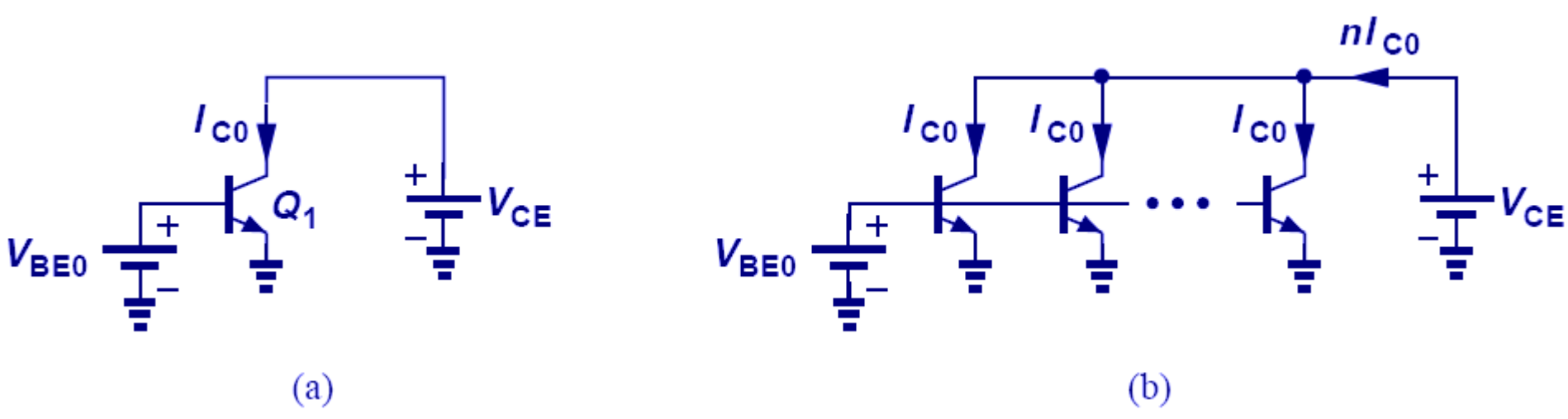
- The value of I_{C0} must be chosen according to the required g_m and, ultimately, the required gain.
- We say the transistor is “biased” at a collector current of I_{C0}, meaning the device carries a bias (or “quiescent”) current of I_{C0} in the absence of signals.



- ✓ g_m can be visualized as the slope of I_C versus V_{BE}.
- ✓ A large I_C has a large slope and therefore a large g_m.

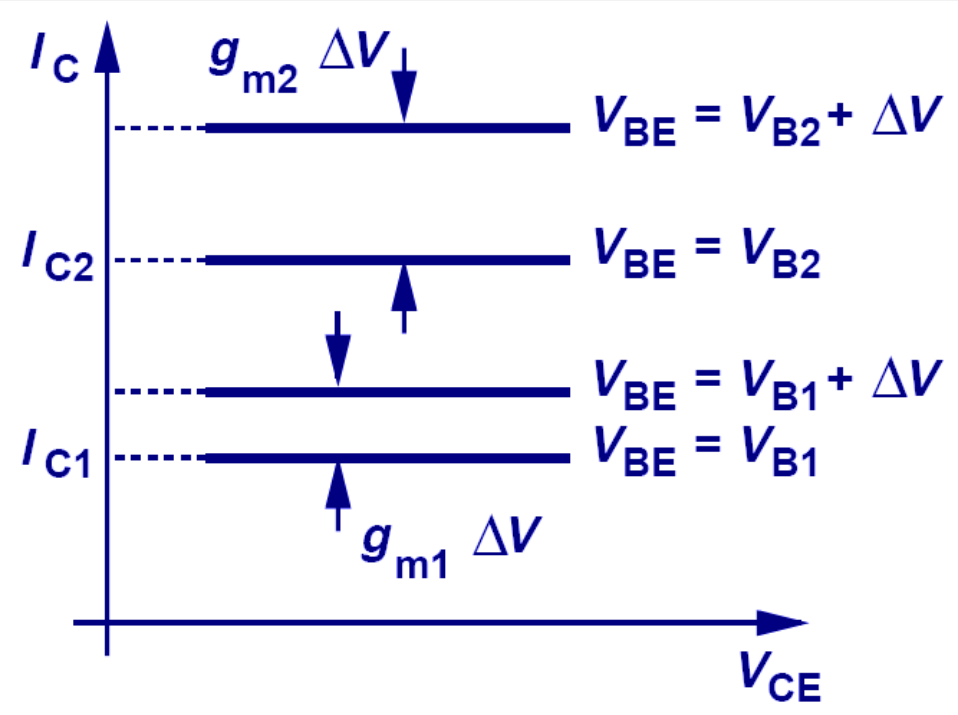
Transconductance and Area

- When the area of a transistor is increased by n, I_S increases by n. For a constant V_{BE}, I_C and hence g_m increases by a factor of n.



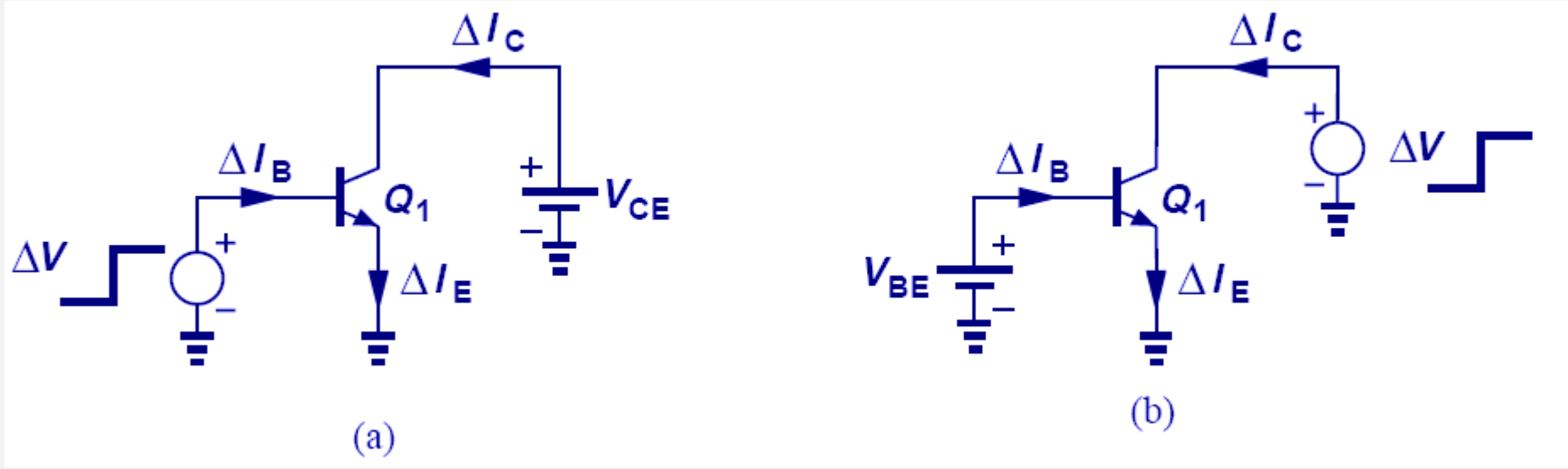
Transconductance and I_c

- For a given V_{BE} swing (ΔV), the current around I_{C2} is larger than it would be around I_{C1}. This is because g_m is larger I_{C2} $(g_{m2} > g_{m1})$.
- The transconductance $(g_m = \frac{I_C}{V_T})$ is fundamentally a function of the collector current rather than the base current.



Small-Signal Model: Derivation

- Small signal model is derived by perturbing voltage difference (ΔV), every two terminals while fixing the third terminal and analyzing the change in current of all three terminals ($\Delta I_C, \Delta I_B, \Delta I_E$).

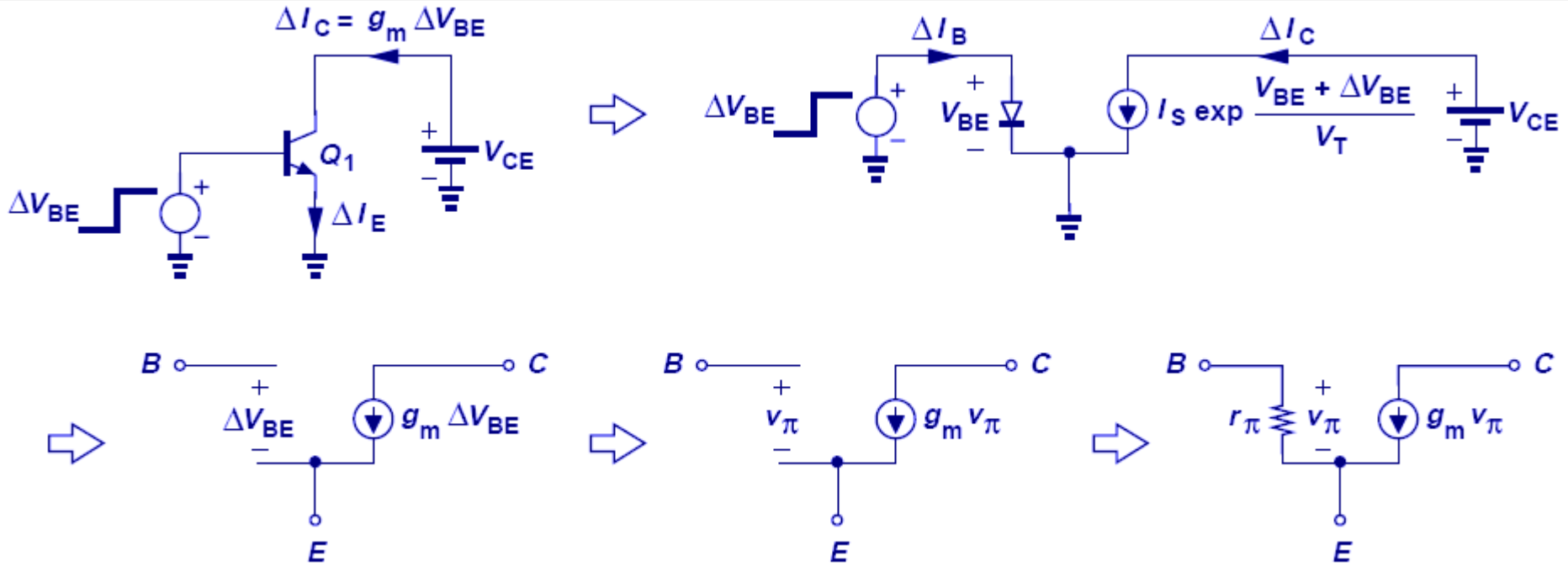


(ΔV_{BE})

(ΔV_{CE})

Small-Signal Model: V_{BE} Change

- Model as a voltage-controlled current source between *CE*: $\Delta I_C = g_m \Delta V_{BE}$.
- Model as a resistor between *BE*: $\Delta V_{BE} = r_\pi \Delta I_B$, where $r_\pi = \frac{\beta}{g_m}$.



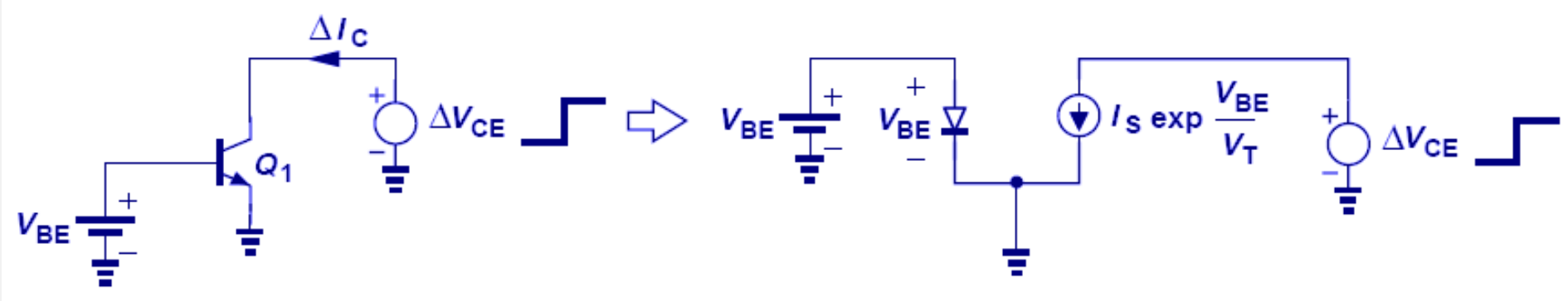
$$\because \Delta I_B = \frac{\Delta I_C}{\beta} = \frac{g_m}{\beta}\Delta V_{BE}$$

$$\therefore \Delta V_{BE} = \frac{\beta}{g_m}\Delta I_B = r_\pi \Delta I_B$$

$(v_\pi \triangleq \Delta V_{BE})$

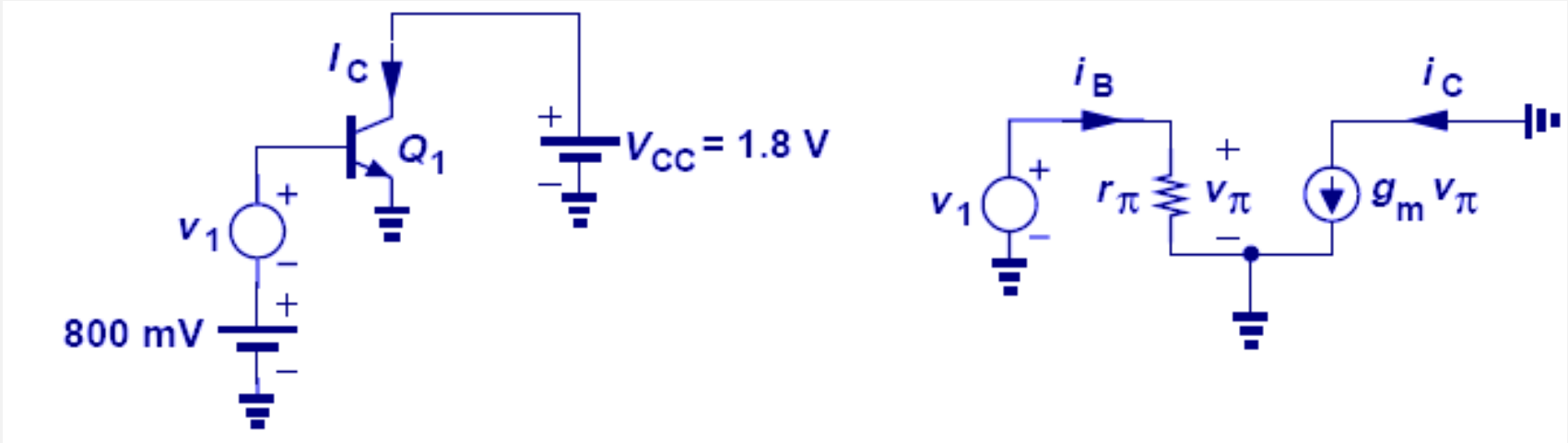
Small-Signal Model: V_{CE} Change

- Ideally, V_{CE} has no effect on the collector current. Thus, it will not contribute to the small signal model.
- It can be shown that V_{CB} has no effect on the small signal model, either.



Example 4-10

- Consider the circuit shown below, where v_1 represents the signal generated by a microphone, $I_S = 3 \times 10^{-16}$ A, β = 100, and Q_1 operates in the active mode, (a) If $v_1 = 0$, determine the small-signal parameters of (b) If the microphone generates a 1-mV signal, how much change is observed in the collector and base currents?



- **Solution**:

(a)

$$I_C = I_S \exp \frac{V_{BE}}{V_T}$$
$$= (3 \times 10^{-16}\ \text{A}) \exp \frac{800\ \text{mA}}{26\ \text{mA}}$$
$$= 6.92\ \text{mA}$$

$$g_m = \frac{I_C}{V_T} = \frac{6.92\ \text{mA}}{26\ \text{mA}} = \frac{1}{3.75\ \Omega}$$

$$r_\pi = \frac{\beta}{g_m} = \frac{100}{(1/3.75\ \Omega)} = 375\ \Omega$$

(b)

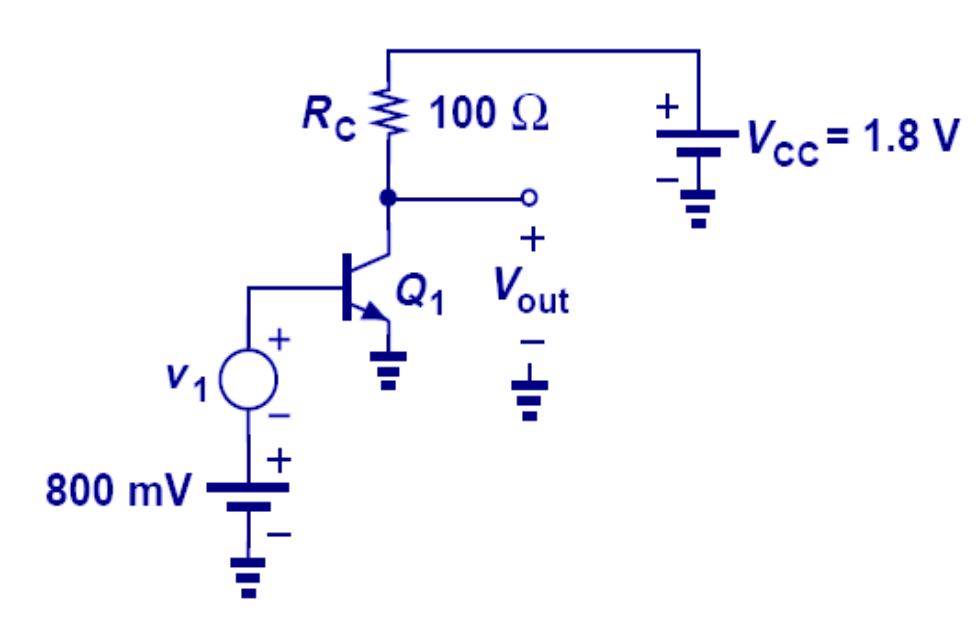
$$\Delta I_C = g_m v_1$$
$$= \frac{1}{3.75\ \Omega} \times 1\ \text{mV}$$
$$= 0.267\ \text{mA}$$

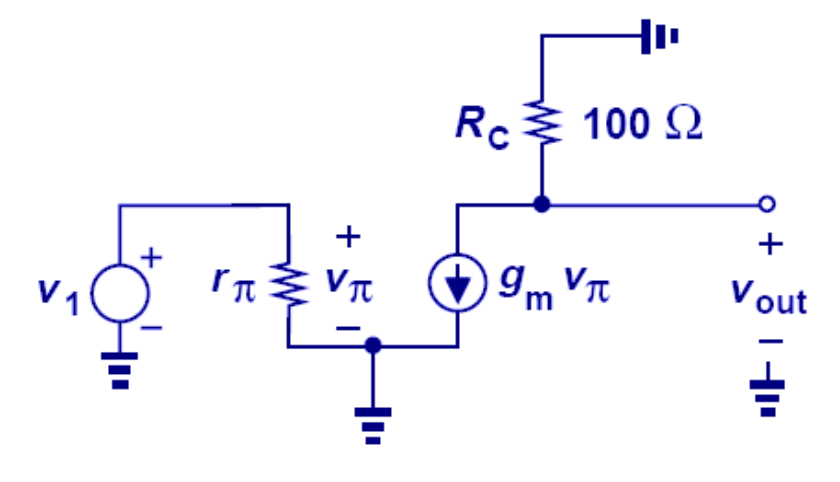
$$\because \Delta I_B = \frac{1}{r_\pi} v_1$$
$$= \frac{1}{375\ \Omega} \times 1\ \text{mV}$$
$$= 2.67\ \mu\text{A}$$

$$\left(\text{Note: } \Delta I_B = \frac{\Delta I_C}{\beta} = \frac{0.267\ \text{mA}}{100}\right.$$
$$\left. = 2.67\ \mu\text{A}\right)$$

Example 4-11

- Consider the circuit shown below, where resistor R_c converts the collector current to a voltage, (a) Verify that the transistor ($I_S = 3 \times 10^{-16}$ A and β = 100) operates in the active mode, (b) Determine the output signal level if the microphone produces a 1-mV signal.





- **Solution**:

(a)

$$I_C = I_S \exp\frac{V_{BE}}{V_T}$$
$$= (3 \times 10^{-16}\ \text{A}) \exp\frac{800\ \text{mA}}{26\ \text{mA}}$$
$$= 6.92\ \text{mA}$$

$$V_{out} = V_{CC} - R_C I_C$$
$$= 1.8 - 100 \times 6.92 \times 10^{-3}$$
$$= 1.108\ \text{V}$$

$(V_{out} = 1108\ \text{mV}) > 800\ \text{mV}$, the B-C diode is reverse-biased.

The Q_1 is in forward active mode.

(b)

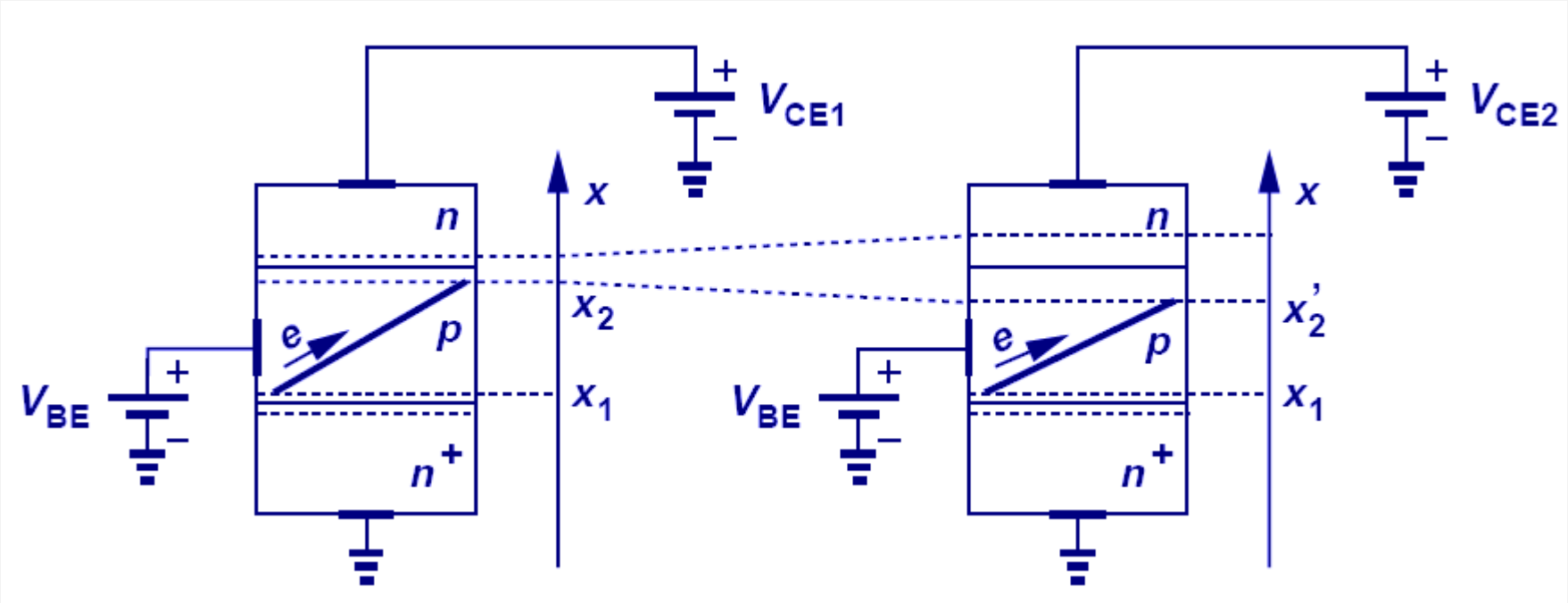
$$g_m = \frac{I_C}{V_T} = \frac{6.92\ \text{mA}}{26\ \text{mA}} = \frac{1}{3.75\ \Omega}$$
$$\Delta I_C = g_m v_1$$
$$= \frac{1}{3.75\ \Omega} \times 1\ \text{mV}$$
$$= 0.267\ \text{mA}$$

$$\because v_{out} = -R_C \Delta I_C$$
$$= -100 \times (0.267 \times 10^{-3})$$
$$= -26.7\ \text{mV}$$

$$\text{Voltage Gain} = \frac{v_{out}}{v_1} = \frac{-26.7\ \text{mV}}{1\ \text{mV}}$$
$$= -26.7$$

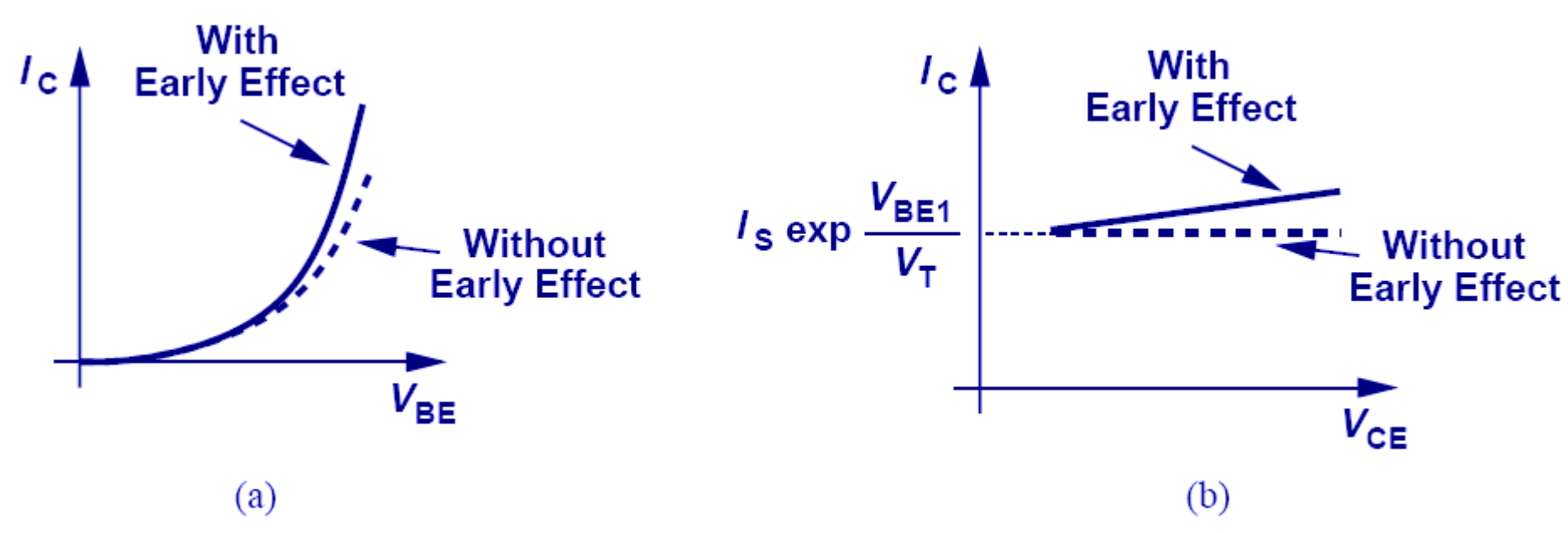
Early Effect

- The claim that collector current does not depend on V_{CE} is not accurate.
- As V_{CE} increases, the depletion region between base and collector increases. Therefore, the effective base width decreases, which leads to an increase in the collector current.



Transistor Model with Early Effect

- With Early effect, collector current becomes larger than usual and a function of V_{CE}.



$$I_C \approx \left(I_S \exp \frac{V_{BE}}{V_T}\right)\left(1 + \frac{V_{CE}}{V_A}\right)$$

where V_A: is called the "Early voltage."

$$\frac{dI_C}{dV_{CE}} = \left(I_S \exp \frac{V_{BE}}{V_T}\right)\left(\frac{1}{V_A}\right) \approx \frac{I_C}{V_A}$$

Example 4-13

- A bipolar transistor carries a collector current of 1 mA with V_{CE} = 2 V. Determine the required base-emitter voltage if $V_A = \infty$ or $V_A = 20$ V. Assume $I_S = 2 \times 10^{-16}$ A.
- **Solution**:

$$I_C = \left(I_S \exp\frac{V_{BE}}{V_T}\right)\left(1 + \frac{V_{CE}}{V_A}\right)$$

$$I_S \exp\frac{V_{BE}}{V_T} = \frac{I_C}{\left(1 + \frac{V_{CE}}{V_A}\right)}$$

$$\exp\frac{V_{BE}}{V_T} = \frac{I_C}{I_S}\frac{1}{1 + \frac{V_{CE}}{V_A}}$$

$$V_{BE} = V_T \ln\left(\frac{I_C}{I_S}\frac{1}{1 + \frac{V_{CE}}{V_A}}\right)$$

$V_A = \infty$,

$$V_{BE} = V_T \ln\left(\frac{1 \times 10^{-3}}{2 \times 10^{-16}}\frac{1}{1 + \frac{2}{\infty}}\right)$$

$$= 760 \text{ mV}$$

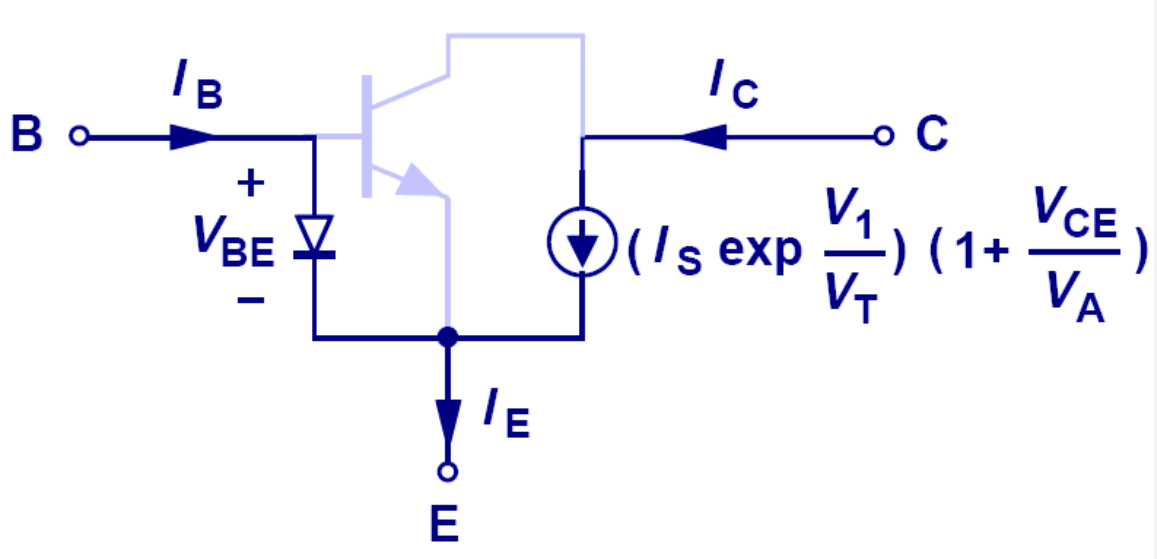
$V_A = 20$,

$$V_{BE} = V_T \ln\left(\frac{1 \times 10^{-3}}{2 \times 10^{-16}}\frac{1}{1 + \frac{2}{20}}\right)$$

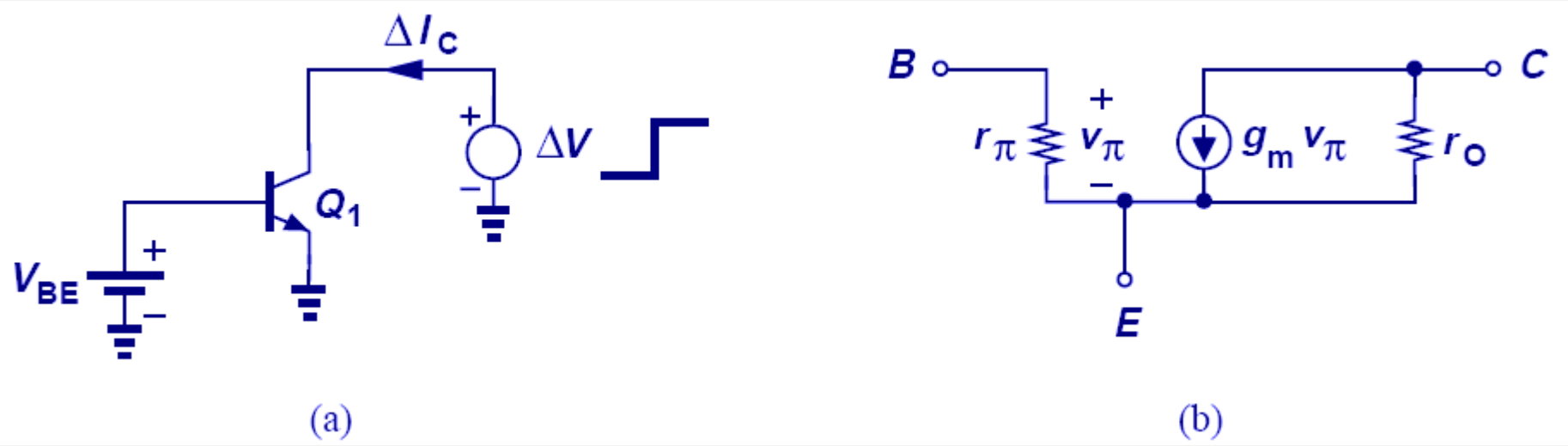
$$= 758 \text{ mV}$$

Early Effect and Large-Signal Model

- Early effect can be accounted for in large-signal model by simply changing the collector current with a correction factor.
- In this mode, base current does not change.



Early Effect and Small-Signal Model



$$r_o = \frac{\Delta V_{CE}}{\Delta I_C} = \frac{V_A}{I_S \exp \frac{V_{BE}}{V_T}} \approx \frac{V_A}{I_C}$$

Example 4-14

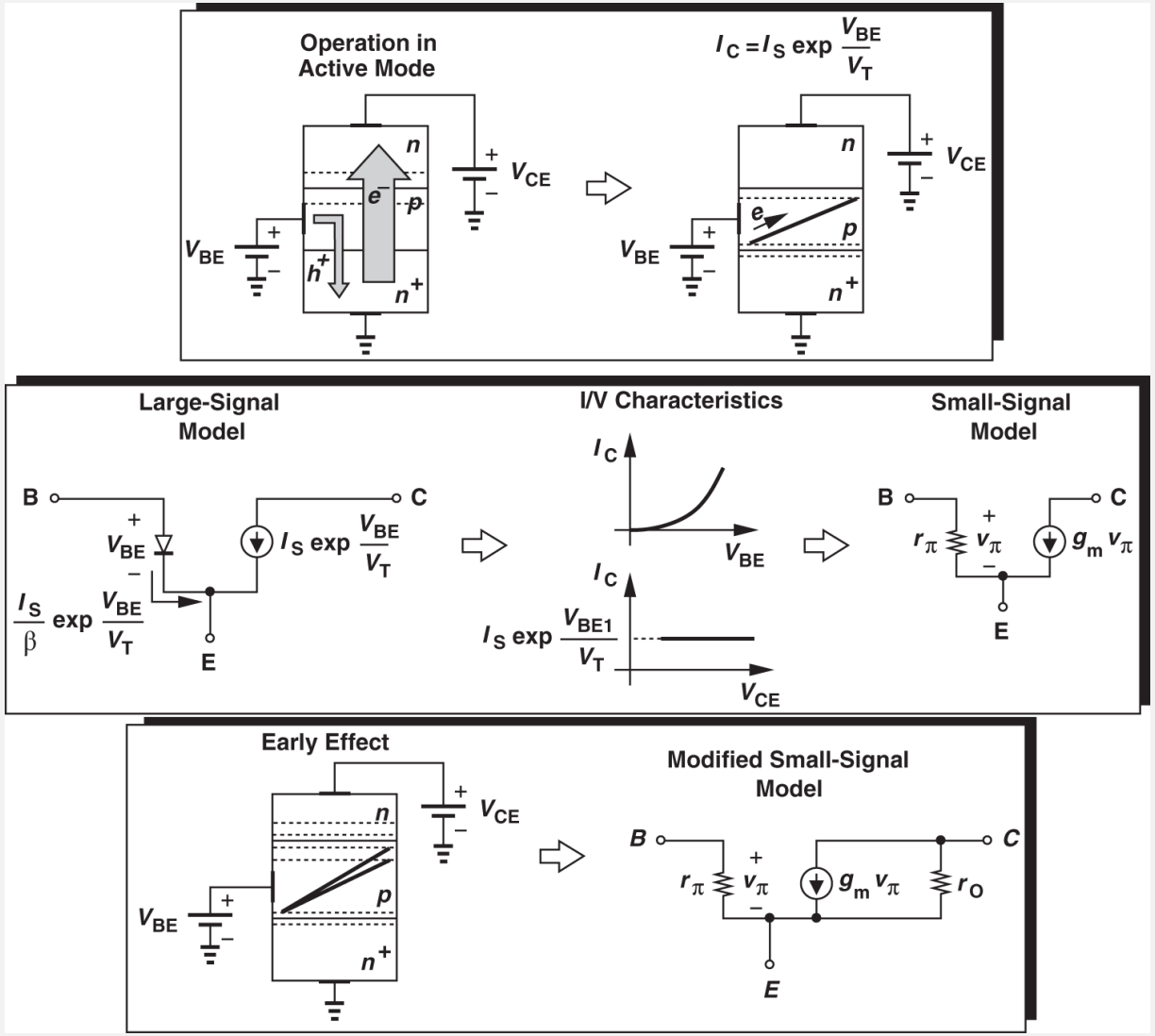
- A transistor is biased at a collector current of 1 mA. Determine the small-signal model if $\beta = 100$ and $V_A = 15$ V.
- **Solution**:

$$g_m = \frac{I_C}{V_T} = \frac{1\text{ mA}}{26\text{ mA}} = \frac{1}{26\ \Omega}$$

$$r_\pi = \frac{\beta}{g_m} = \frac{100}{(1/26\ \Omega)} = 2.6\ \text{k}\Omega$$

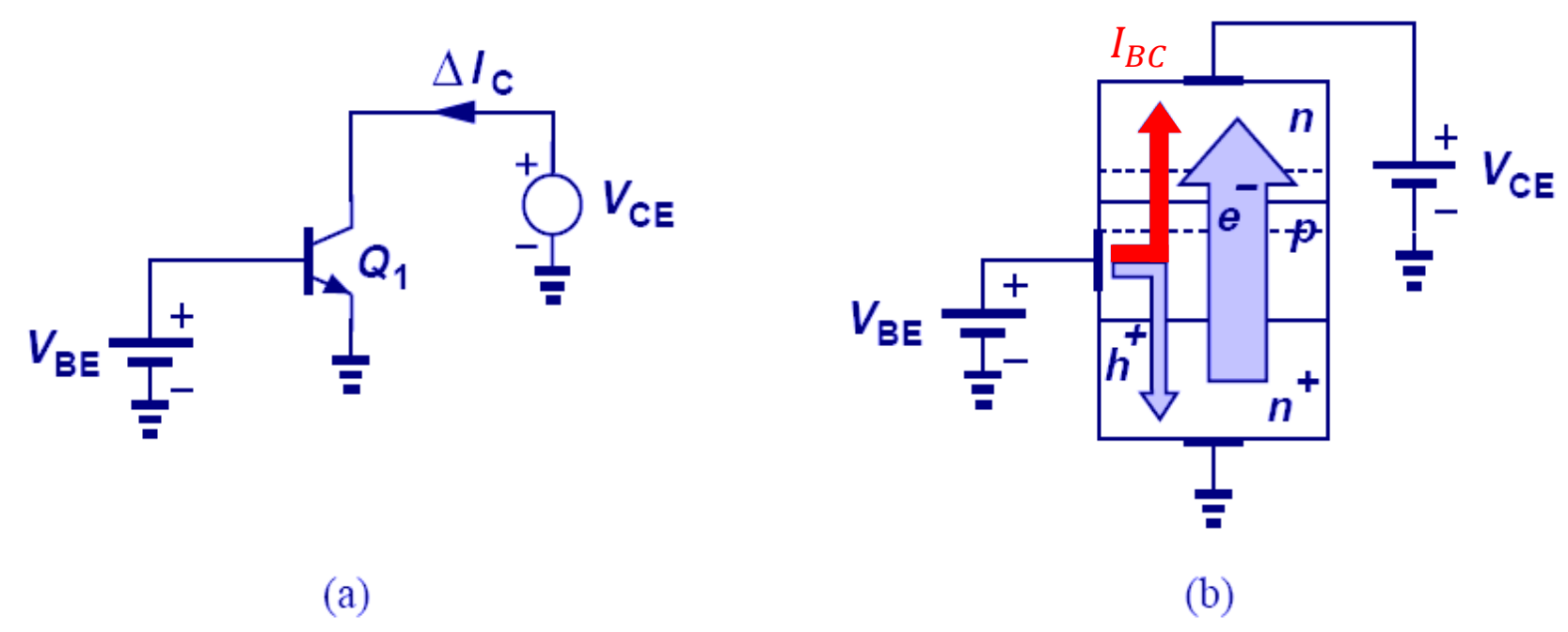
$$r_o = \frac{V_A}{I_C} = \frac{15}{1 \times 10^{-3}} = 15\ \text{k}\Omega$$

Summary


Operation in Active Mode
$I_C = I_S \exp \frac{V_{BE}}{V_T}$
n
p
n^+
V_{CE}
V_{BE}
e^-
h^+
Large-Signal Model
I/V Characteristics
Small-Signal Model
B
C
E
$\frac{I_S}{\beta} \exp \frac{V_{BE}}{V_T}$
$I_S \exp \frac{V_{BE}}{V_T}$
I_C
V_{BE}
V_{CE}
$I_S \exp \frac{V_{BE1}}{V_T}$
r_π
v_π
$g_m v_\pi$
Early Effect
Modified Small-Signal Model
r_O

Bipolar Transistor in Saturation

- When collector voltage drops below base voltage and forward biases the collector-base junction, base current increases and decreases the current gain factor, β.



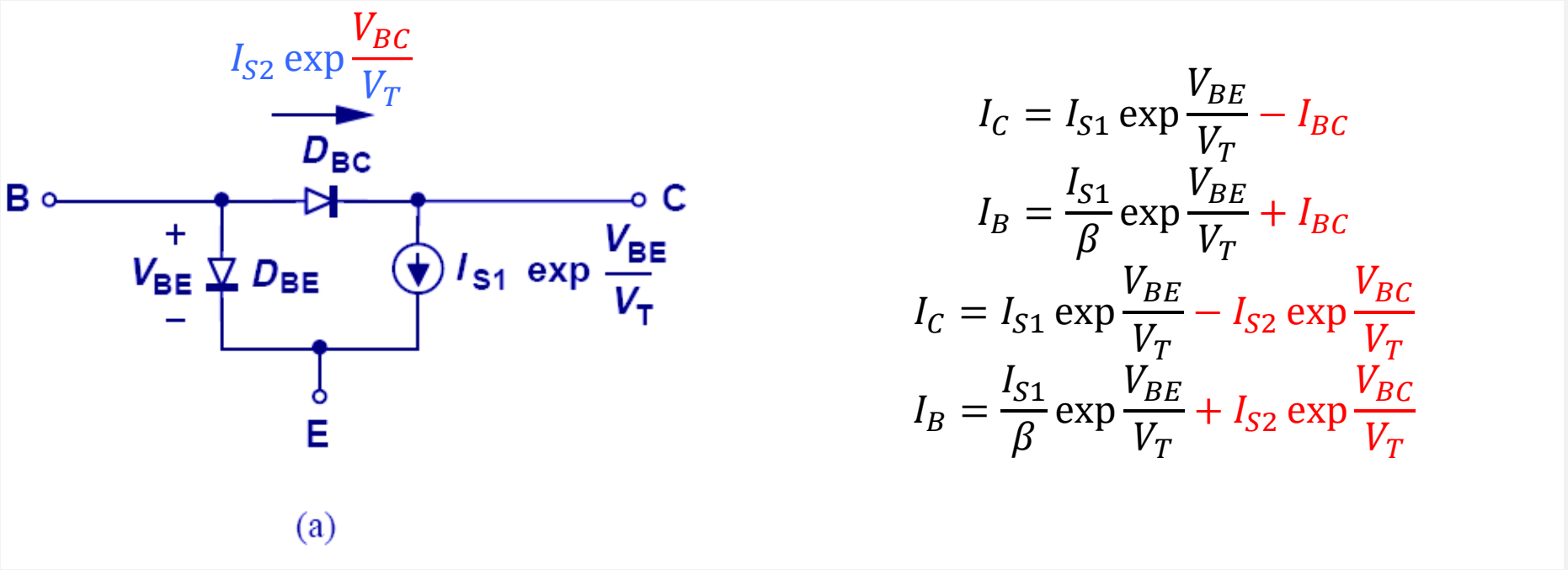
$$I_C = I_{S1} \exp \frac{V_{BE}}{V_T} - I_{BC}$$

$$I_B = \frac{I_{S1}}{\beta} \exp \frac{V_{BE}}{V_T} + I_{BC}$$

$$\beta_{forced} \triangleq \frac{I_C}{I_B} < \beta$$

Large-Signal Model for Saturation Region

- Note that the net collector current decreases as the device enters saturation because the B-C diode is forward-biased.



$$I_C = I_{S1} \exp \frac{V_{BE}}{V_T} - I_{BC}$$

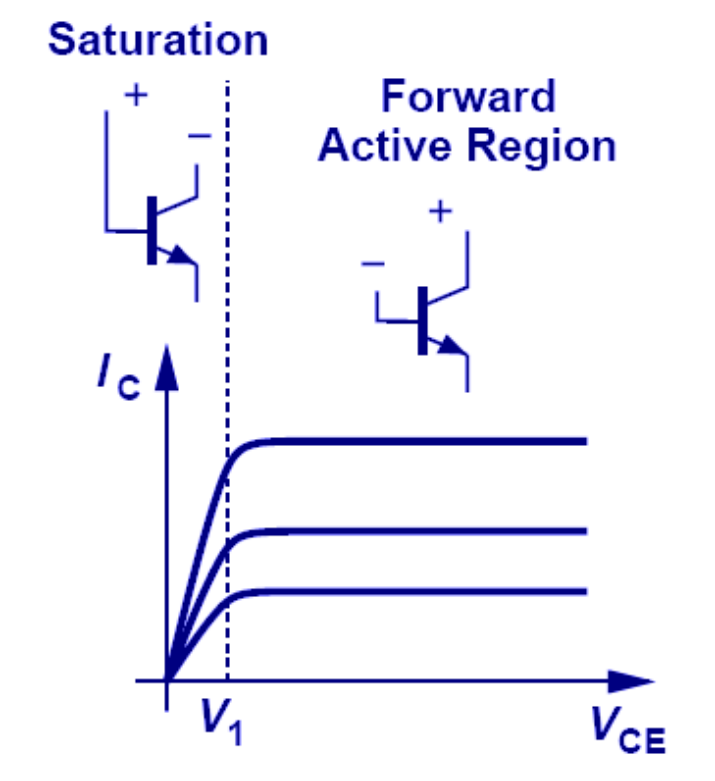
$$I_B = \frac{I_{S1}}{\beta} \exp \frac{V_{BE}}{V_T} + I_{BC}$$

$$I_C = I_{S1} \exp \frac{V_{BE}}{V_T} - I_{S2} \exp \frac{V_{BC}}{V_T}$$

$$I_B = \frac{I_{S1}}{\beta} \exp \frac{V_{BE}}{V_T} + I_{S2} \exp \frac{V_{BC}}{V_T}$$

Overall I/V Characteristics

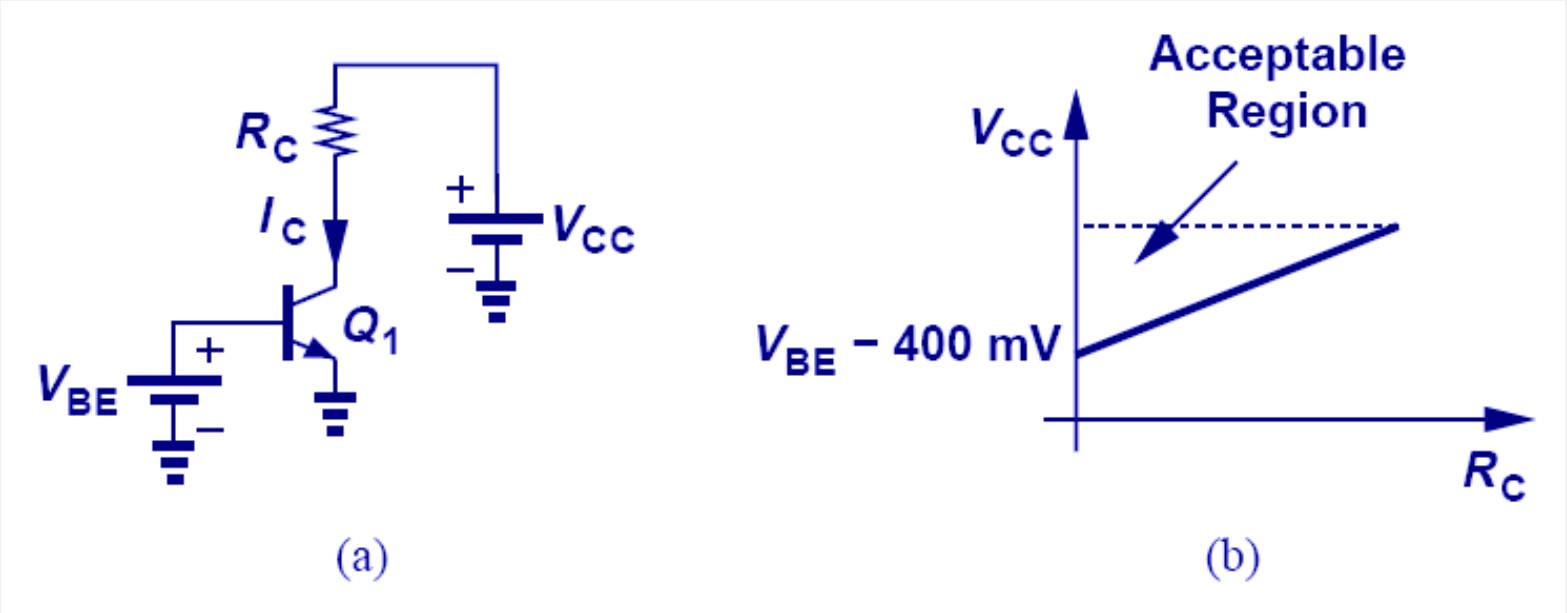
- Ic begins to fall for V_{CE} less than V_1. The term "saturation" is used because increasing the base current in this region of operation leads to little change in the collector current.
- In addition to a drop in β, the speed of bipolar transistors also degrades in saturation.



As a rule of thumb, we permit soft saturation with V_{BC} < 400 mV because the current in the B-C junction is negligible, provided that various tolerances in the component values do not drive the device into deep saturation.

Example 4-16

- Determine the relation between R_C and V_{CC} that guarantees operation in soft saturation or active region.



- **Solution**: In soft saturation, the collector current is still equal to $I_S \exp \frac{V_{BE}}{V_T}$. The collector voltage must not fall below the base voltage by more than 400 mV:

$$V_{BC} \leq 400 \text{ mV}$$

$$V_{BE} - V_{CE} \leq 400 \text{ mV}$$

$$V_{CE} \geq \underbrace{V_{BE} - 400 \text{ mV}}_{=V_1}$$

$$V_{CC} - R_C I_C \geq V_{BE} - 400 \text{ mV}$$

$$V_{CC} \geq I_C R_C + (V_{BE} - 400 \text{ mV})$$

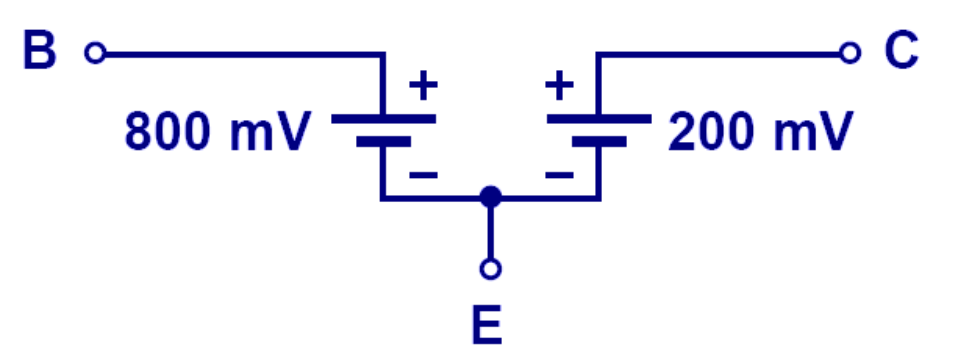
Given a V_{BE}, $I_C = I_S \exp \frac{V_{BE}}{V_T}$ is a constant. The lower bound of V_{CC} is a straight line

$$V_{CC} = mR_C + b,$$

where $m = I_C$ and $b = V_{BE} - 400$ mV.

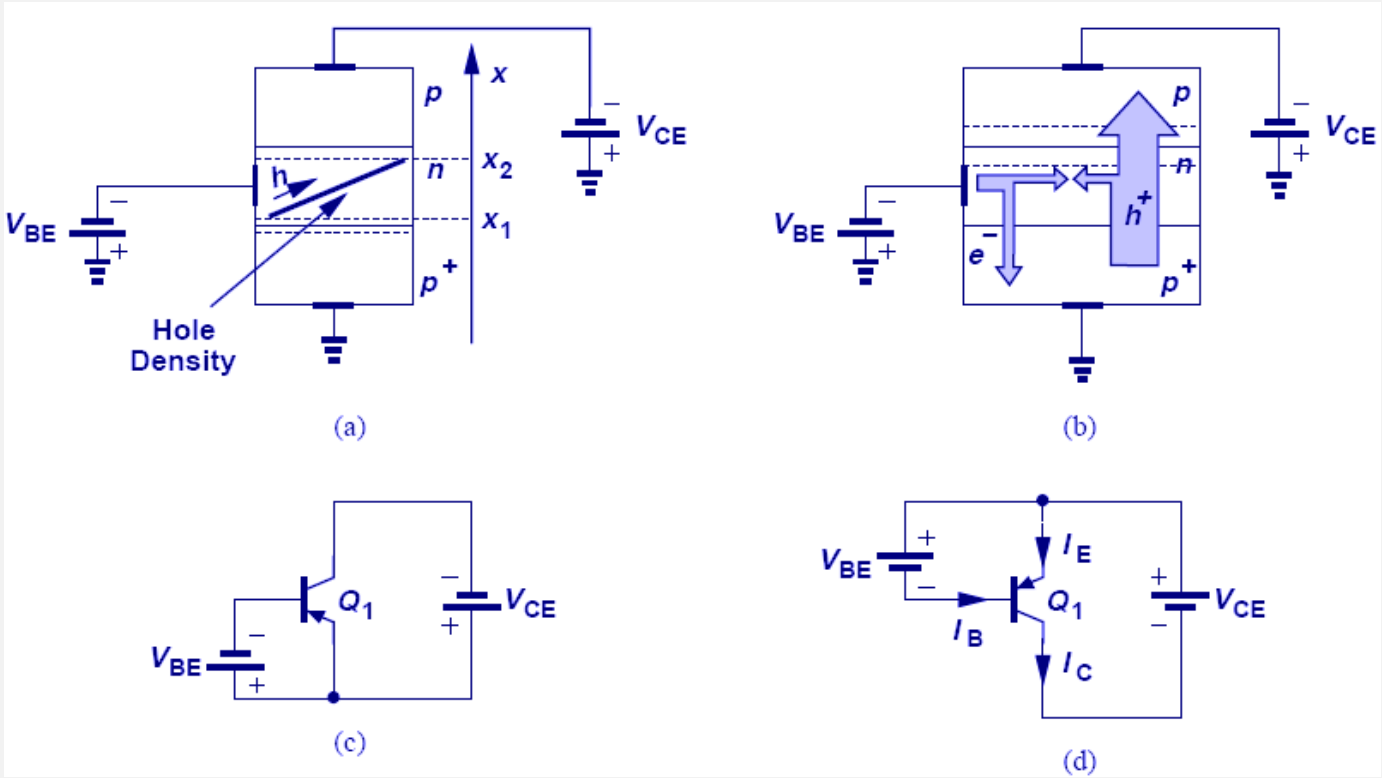
Deep Saturation

- In deep saturation region, the transistor loses its voltage-controlled current capability and V_{CE} becomes constant $(V_{CE,sat} \approx 200\ \mathrm{mV})$.



PNP Transistor

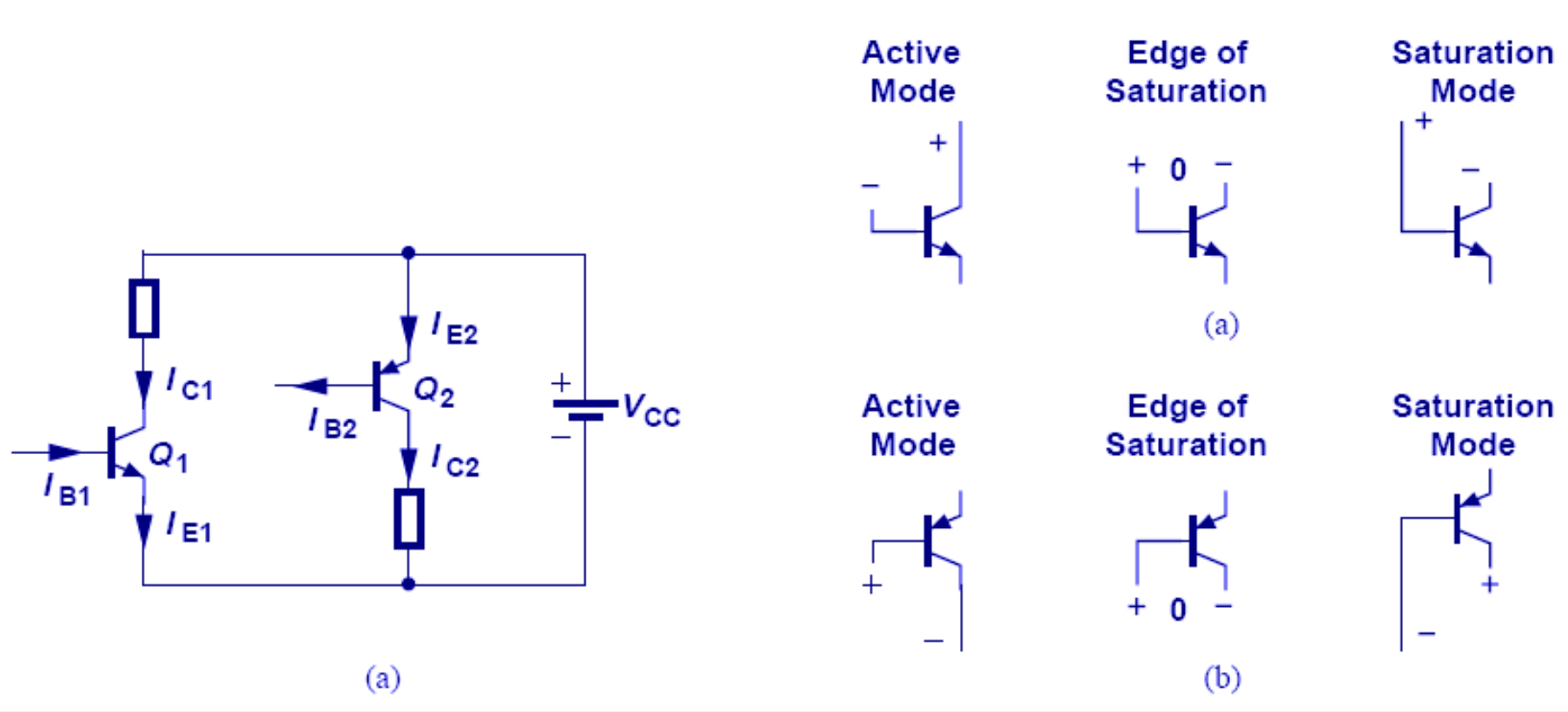
- With the polarities of emitter, collector, and base reversed, a PNP transistor is formed.
- All the principles that applied to NPN's also apply to PNP's, with the exception that emitter is at a higher potential than base and base at a higher potential than collector.



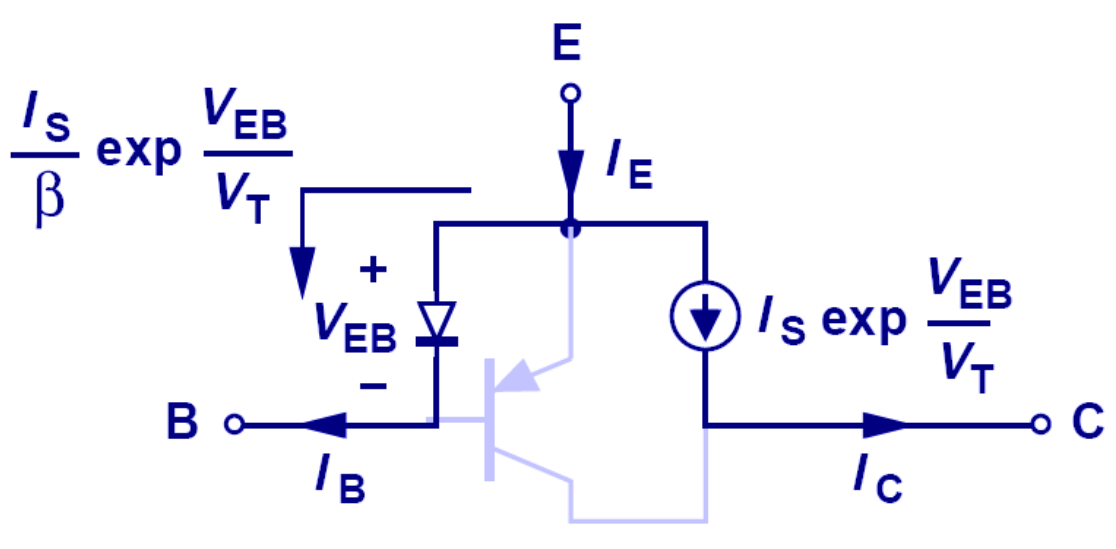
(a) Structure of pnp transistor,
(b) current flow in pnp transistor,
(c) proper biasing,
(d) more intuitive view of (c).

A Comparison between NPN and PNP Transistors

- The figure below summarizes the direction of current flow and operation regions for both the NPN and PNP BJT's.



Large Signal Model for PNP



$$I_C = I_S \exp\frac{V_{EB}}{V_T}$$

$$I_B = \frac{I_C}{\beta}$$

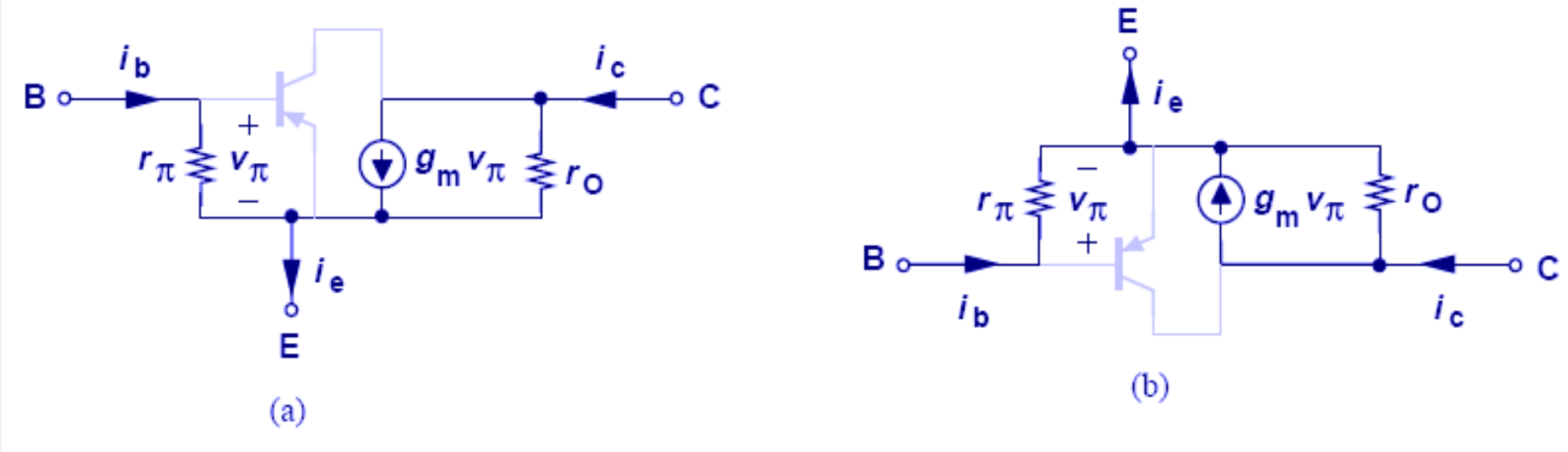
$$I_E = \frac{\beta + 1}{\beta} I_C$$

With Early effect,

$$I_C = \left(I_S \exp\frac{V_{EB}}{V_T}\right)\left(1 + \frac{V_{EC}}{V_A}\right)$$

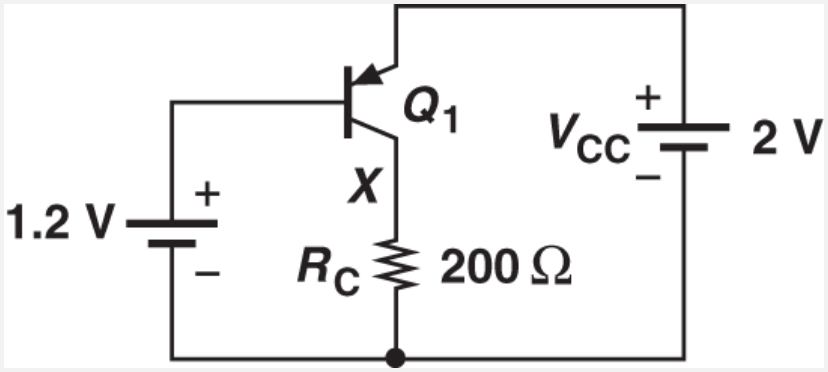
Small-Signal Model for PNP Transistor

- The small signal model for PNP transistor is exactly IDENTICAL to that of NPN. This is not a mistake because the current direction is taken care of by the polarity of $V_{BE.}$



Example 4-17

- Determine the terminal currents of Q_1 and verify operation in the forward active region. Assume $I_S = 2 \times 10^{-16}$ A and $\beta = 50$ and $V_A = \infty$.



- **Solution**:

$$I_C = \left(I_S \exp\frac{V_{EB}}{V_T}\right)\left(1 + \frac{V_{EC}}{V_A}\right)$$

$$\because V_A = \infty, I_C = I_S e^{V_{EB}/V_T}$$

$$I_C = 2 \times 10^{-16} e^{(2-1.2)/0.26} = 4.61\ \text{mA}$$

$$I_B = \frac{I_C}{\beta} = \frac{4.61\ \text{mA}}{50} = 92.2\ \mu\text{A}$$

$$I_E = \frac{\beta + 1}{\beta} I_C = \frac{50 + 1}{50}(4.61\ \text{mA}) = 4.70\ \text{mA}$$

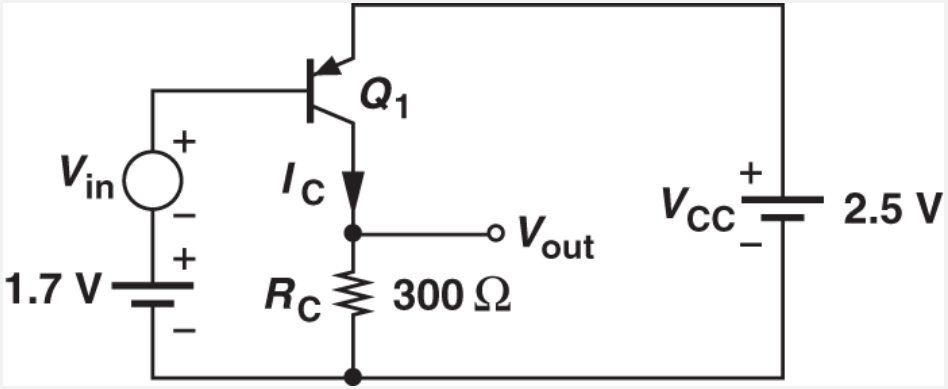
$$V_X = I_C R_C = 4.61\ \text{mA} \times 200 = 0.922\ \text{V}$$

$(V_B = 1.2) > (V_C = V_X = 0.922)$, the B-C diode is reverse-biased.

The Q_1 is in forward active mode.

Example 4-18

- V_{in} represents a signal generated by a microphone. Determine V_{out} for V_{in} = 0 and V_{in} = +5 mV if $I_S = 1.5 \times 10^{-16}$ A.



- **Solution**:

For $V_{in} = 0$,

$$I_C = I_S e^{V_{EB}/V_T}$$

$$I_C = 1.5 \times 10^{-16} e^{(2.5-1.7)/0.26} = 3.46\ \text{mA}$$

$$V_{out} = I_C R_C = 3.46\ \text{mA} \times 300 = 1.038\ \text{V}$$

$(V_B = 1.7) > (V_C = V_{out} = 1.04)$, the B-C diode is reverse-biased.

The Q_1 is in forward active mode.

For $V_{in} = 5\ \text{mV}$,

$$I_C = I_S e^{V_{EB}/V_T}$$

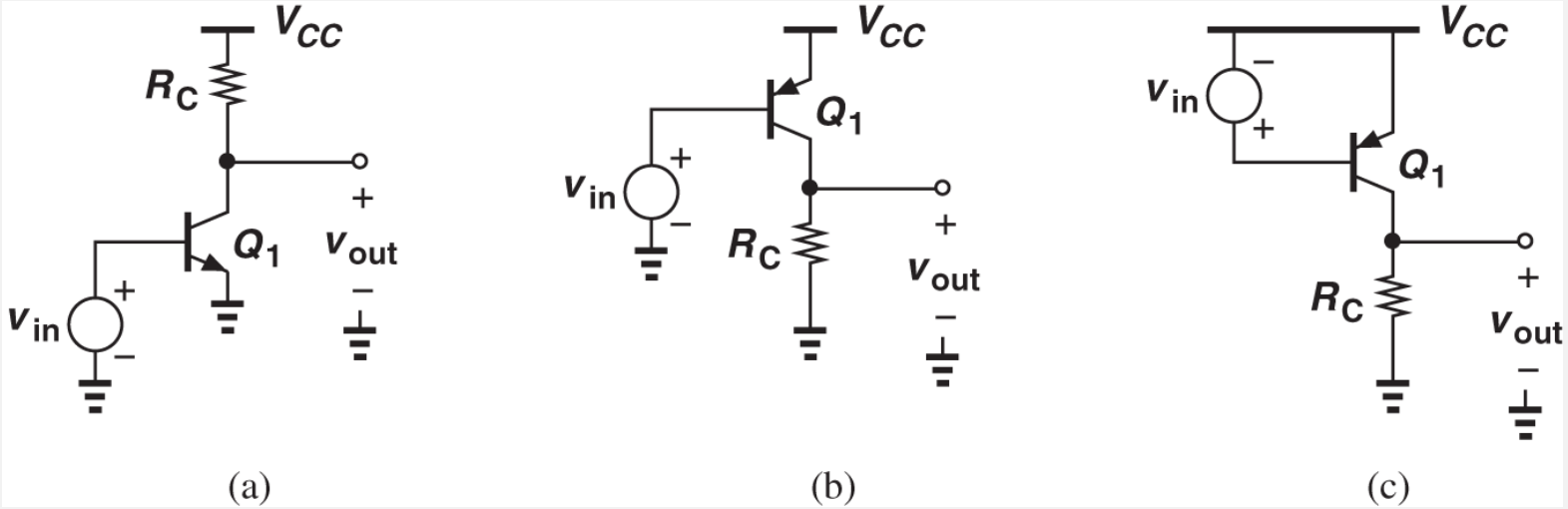
$$I_C = 1.5 \times 10^{-16} e^{(2.5-1.705)/0.26} = 2.85\ \text{mA}$$

$$V_{out} = I_C R_C = 2.85\ \text{mA} \times 300 = 0.856\ \text{V}$$

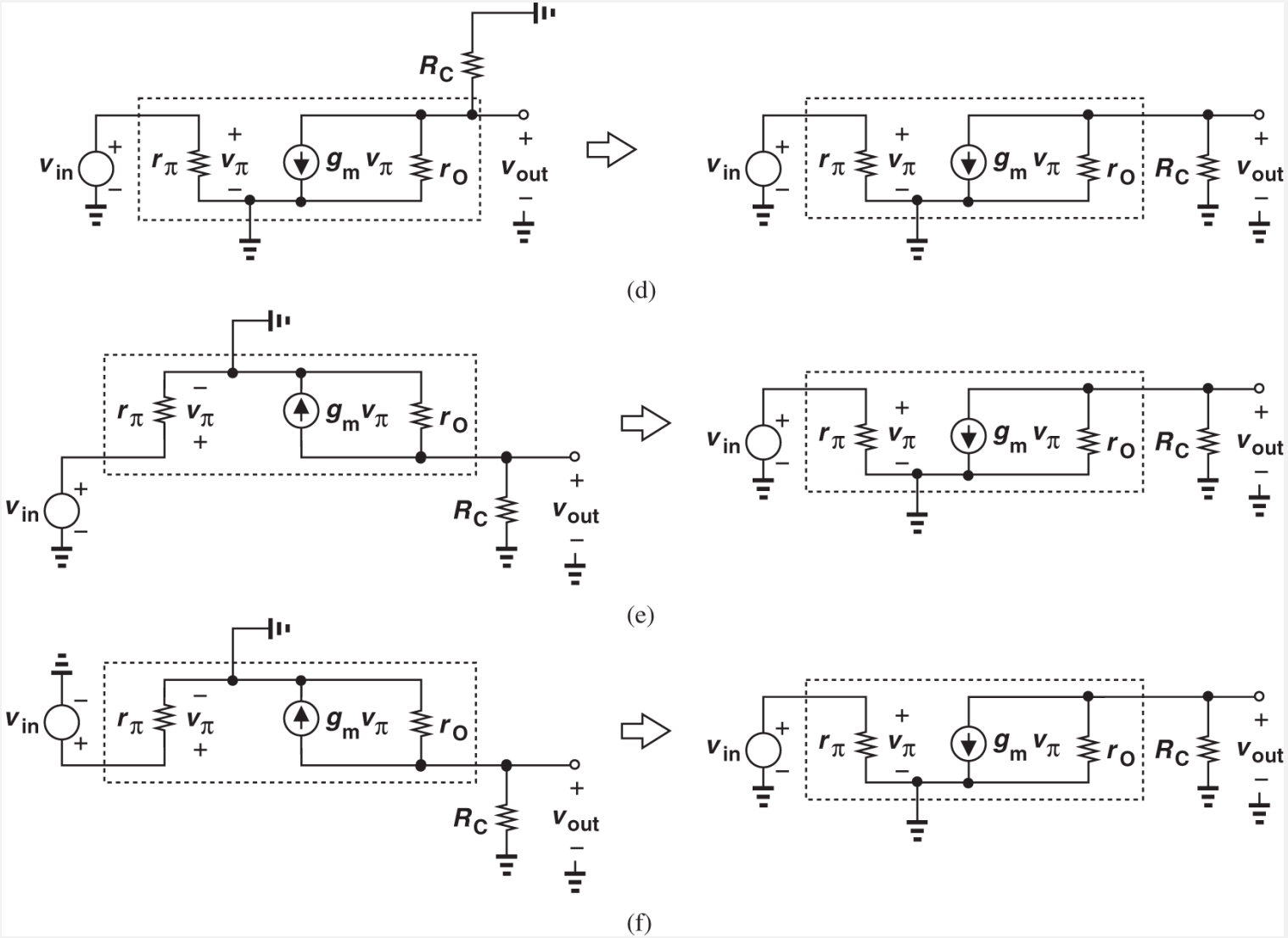
$$\text{Voltage gain} = \frac{\Delta V_{out}}{\Delta V_{in}} = \frac{(0.856 - 1.038)}{(0.005 - 0)} = -36.4$$

Example 4-20

- Draw the small-signal equivalent circuits for the topologies shown below in Figs. (a)-(c) and compare the results.



- **Solution**:



All three topologies reduce to the same equivalent circuit because Vcc is grounded in the small-signal representation.